

GATITOS: Using a New Multilingual Lexicon for Low-resource Machine Translation

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Abstract

Modern machine translation models and language models are able to translate without having been trained on parallel data, greatly expanding the set of languages that they can serve. However, these models still struggle in a variety of predictable ways, a problem that cannot be overcome without at least some trusted bilingual data. This work expands on a cheap and abundant resource to combat this problem: bilingual lexica (BiLEXs). We test the efficacy of bilingual lexica in a real-world setup, on 200-language translation models trained on web-crawled text. We present several findings: (1) using lexical data augmentation, we demonstrate sizable performance gains for unsupervised translation; (2) we compare several families of data augmentation, demonstrating that they yield similar improvements, and can be combined for even greater improvements; (3) we demonstrate the importance of carefully curated lexica over larger, noisier ones, especially with larger models; and (4) we compare the efficacy of multilingual lexicon data versus human-translated parallel data. Based on results from (3), we develop and open-source GATITOS, a high-quality, curated dataset covering 170 mostly low-resource languages at the time of this submission, one of the first human-translated resources to support many of these languages¹.

1 Introduction

Neural machine translation (NMT) has emerged as the dominant way of training machine translation models (Bahdanau et al., 2015), where translation is modeled as a sequence-to-sequence task to be learned by neural networks (Sutskever et al., 2014). Massively multilingual machine translation

(MMMT) refers to the concept of training a single machine translation model on many languages and language pairs using a shared set of parameters, and has also seen success in recent years (Firat et al., 2016; Wu et al., 2016; Johnson et al., 2017; Aharoni et al., 2019; Fan et al., 2022; NLLBTeam et al., 2022; Bapna et al., 2022; Siddhant et al., 2022). Training these models typically relies on large-scale parallel corpora mined from the web (Resnik and Smith, 2003; Uszkoreit et al., 2010; Esplà-Gomis, 2009; Bañón et al., 2020).

However, beyond the traditional technique of training NMT models with human-translated parallel texts, a number of other strategies have shown success recently, especially on lower-resource languages. One of these techniques is self-supervised training using monolingual corpora (Siddhant et al., 2020; Cheng et al., 2021). With this approach, NMT models are pretrained or jointly trained on a self-supervised task with monolingual data, such as the MASS (Song et al., 2019b) or BART (Lewis et al., 2020; Liu et al., 2020) tasks, as well as the usual neural machine translation task. This training regime can aid the model in performing zero-shot translation (Bapna et al., 2022; Siddhant et al., 2022), in cases where a language has monolingual data but no parallel data. Moreover, both the self-supervised task and the supervised MT task can be modeled as neural sequence-to-sequence (Seq2Seq) problems, meaning a single Seq2Seq model can be used for training on both tasks.

Other techniques that have proven useful for low-resource MT include back-translation (Sennrich et al., 2016; Caswell et al., 2019; Feldman and Coto-Solano, 2020) and the incorporation of language models into MT training (Gulcehre et al., 2017; Baziotis et al., 2020; Freitag et al., 2022b). There has also been extensive work on training completely unsupervised MT systems using monolingual corpora only (Artetxe et al., 2017, 2019). For example, Artetxe et al. (2017) uses a

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¹<https://github.com/google-research/url-nlp/tree/main/gatitos>

combination of denoising autoencoding with pre-trained cross-lingual embeddings and on-the-fly back-translation to achieve reasonable MT performance with zero parallel data.

In our work, we supplement the approach that combines supervised and self-supervised training with multilingual lexica. The motivation for using this resource is as follows. Despite the successes of the approach combining supervised and self-supervised training, cross-lingual vocabulary alignment is still highly imperfect in these models, especially for low-resource and unsupervised languages (see [Bapna et al. \(2022\)](#) for examples of some common failure modes). That is, training on all languages using a shared set of parameters is insufficient to induce perfect cross-lingual vocabulary alignment. Of course, we are not the first to experiment with multilingual lexica to improve NMT performance, or multilingual NLP applications more generally; Section 2 gives more details.

Using the publicly available massively multilingual lexicon Panlex ([Kamholz et al., 2014](#)), we demonstrate that this added lexical data leads to small but significant gains over a baseline model on average, even for high-resource languages; and with smaller but carefully curated bilingual lexica, the gains are substantially larger. In both cases, the gains are most significant for unsupervised and low-resource languages. Our contributions are as follows:

1. We provide a thorough comparison of several lexicon-based data augmentation variants for MT, all of which are simple, generalizable, and easy to implement;
2. We test these approaches “in the wild”, i.e. on in a highly multilingual, web-mined data regime such as production systems tend to use, with hundreds of languages and billions of monolingual and parallel sentences;
3. We explore the effects of lexical data quality and quantity;
4. We demonstrate the efficacy of bilingual lexicon-based approaches as models scale;
5. We open-source the high-quality multilingual GATITOS lexicon for low-resource languages.

The **tl;dr** of this paper is that bilingual lexica help low-resource and zero-shot NMT in almost all cases, and that most training-time augmentation methods have similar efficacy, and can be combined to be more effective. When scaling up to larger and more expressive models, these methods

retain their efficacy, but the quality of the translated bilingual lexica becomes more important than the sheer quantity of lexical data points used for data augmentation. For instance, small, high-quality lexica like GATITOS show about 5x larger CHRF improvement than larger, noisier lexica like Panlex.

Throughout this paper, experiments are done on only 24 GATITOS languages; based on the success of these, we expand the dataset to 170 languages, all low-resource, and open-source it.

2 Related Work

A number of works have looked at using multilingual lexicon data augmentation for NMT and other NLP tasks. The first class of augmentations that we experiment with is “codeswitching,” where words in the source sentence are swapped out for their dictionary translations to create mixed-language sentences. This approach has been used for a range of multilingual NLP tasks, including MT ([Reid and Artetxe, 2022](#); [Yang et al., 2020](#); [Liu et al., 2021](#); [Lin et al., 2020, 2021](#); [Pan et al., 2021](#); [Yang et al., 2021](#); [Kumar et al., 2022](#); [Khatri et al., 2021](#); [Kuwanto et al., 2021](#); [Xia et al., 2019](#)). Many of these, however, only look at codeswitching between the source and target languages, e.g. substituting source words with dictionary translations into the target language, or word-for-word BiLex translations of the target to make synthetic back-translated data ([Nag et al., 2020](#)). [Qin et al. \(2020\)](#) experiment with codeswitching on NLI, sentiment classification, document classification, dialogue state tracking, and spoken language understanding, [Malon \(2021\)](#) looks at codeswitching embeddings for language modeling, and [Wang et al. \(2022\)](#) experiment on NER, POS tagging, and dependency parsing. Another similar work is [Chaudhary et al. \(2020\)](#), in which the MLM task is modified such that instead of predicting masked source tokens in the *source* language, the authors provided language embeddings to cue the model to predict the masked tokens in a *different* language instead. Codeswitching augmentations go by a variety of different names, e.g. “dictionary denoising” ([Reid and Artetxe, 2022](#)), “Random Aligned Substitution” ([Lin et al., 2020](#)), or “code-mixing”. In our paper, we will stick to the term “codeswitching,” though we will try to point out where an identical or similar approach has been tried under a different name.

The second class of augmentations we experiment with involves prepending lexicon translations

to source sentences as additional cross-lingual signal, as instead of swapping out words in the source sentence. This approach has been tried as well for enhancing MT performance, e.g. in Song et al. (2019a); Maheshwari et al. (2022); Niehues (2021); Michon et al. (2020); Zhong and Chiang (2020); Susanto et al. (2020), and for similar tasks like language modeling (Yu et al., 2021). One potential advantage this approach has over the codeswitching method is that it can be applied at inference time as well: multilingual lexicon entries can be prepended to sentence queries to steer the model toward more accurate word translations. Outside of NMT models, this lexical prompting approach has also been applied to translation with LLMs: Ghazvininejad et al. (2023) provide LLMs with dictionary translations of some of the source sentence words, which the model can use to cover gaps in its vocabulary coverage (although the authors do not experiment with truly low-resource languages). With the rise in popularity of LLMs for MT and other tasks, this is an exciting area for further research.

3 Training data

Models are trained on sentence-level web text.

3.1 Monolingual data

The monolingual training data is from a clean, sentence-level web-mine following the approach set forth in Caswell et al. (2020). To make rapid training and development possible, we subsampled the monolingual data for the 100 highest-resource languages to 10% of its original size. In sum, this totaled about 4B sentences, or about 80B tokens. For the large model experiments in section 8.1, we used the full data, totaling 27B sentences (540B tokens).

3.2 Parallel data

We use an in-house mine of parallel data. It tends to be much noisier than the monolingual data. All parallel data are sampled to 10% of their original size, resulting in 9B parallel sentences (162M tokens) into English and the same number out of English, as well as 700K non-English-centric sentence-pairs. The large models in Section 8.1 use the full dataset.

3.3 Multilingual lexica

3.4 GATITOS

The GATITOS dataset is a new dataset open-sourced in this paper. It consists of 4000 short English seg-

ments translated into 170 very low-resource languages; however, at the time of the experiments in the paper, it covered only 24. The English source text is a mixture of words from a variety of sources, including frequent tokens in the English language, words for numbers, months, days of the week, Swadesh words, names of the languages themselves (including the endonym), and a few short sentences. The tokens were manually reviewed by the authors of this paper to ensure they looked reasonable. As the name implies, this dataset is mostly very short, consisting of 93% single tokens. There are also some short sentences, though only 23 entries have over 5 tokens. We hope this dataset will complement existing publicly available multilingual lexicons like MUSE (Conneau et al., 2017; Lample et al., 2017)

3.5 Panlex

Panlex (Kamholz et al., 2014) is a free, open-access massive online database consisting of word and phrase translations for 5000+ languages, sourced from 2500 individual dictionaries. Panlex contains ≈ 1.3 B translations across all language pairs. For our experiments, we use a subset of the Panlex database covering 177 languages and containing 66M word pairs. Languages were chosen largely by availability of eval sets; details in Appendix J.1.

4 Evaluation

We use two translation evaluation sets: FLORES-200 (NLLBTeam et al., 2022; Goyal et al., 2021; Guzmán et al., 2019), an open-sourced evaluation set consisting of 2009 English Wikipedia sentences translated by humans into 200 languages, and GATONES, an in-house evaluation set of 1,200 English sentences translated into various languages. We use the SacreBLEU (Post, 2018) implementation of CHRF² for our evaluation metric. Higher-quality, embedding-based metrics like BLEURT (Sellam et al., 2020) are not available for these languages, and CHRF seems to be the one of best of the surface-level metrics for low-resource languages (Bapna et al., 2022; Kocmi et al., 2021; Freitag et al., 2022a).

For this study, we only evaluate on English-centric directions. The reason is that, although both evaluation sets are multi-way parallel, they are both also English-original, and thus lose the same infor-

²signature nrefs:1|case:mixed|eff:yes|nc:6|nw:0|space:no|version:2.3.1

mation that pivot translations do. Furthermore, this study places more weight on the $\text{en} \rightarrow \text{xx}$ direction than the $\text{xx} \rightarrow \text{en}$ direction. The main reason for this is that the $\text{xx} \rightarrow \text{en}$ direction is generally an easier direction for models to learn (since they see so much more English text), so the $\text{en} \rightarrow \text{xx}$ direction is usually the limiting reagent when it comes to model quality; as a result we care more about improving this direction.

5 Model

For our experiments we use a Transformer Big encoder-decoder model (Vaswani et al., 2017) with approximately 475M parameters. We train each model for 400K steps on 64 TPU v2 chips. Our models assign a 40% weight to the translation task, and a 60% weight to the MASS task. For models augmented with a monolingual data augmentation, we split the 60% weight on the MASS task into a 30% weight on the augmented data and a 30% weight on the non-augmented data. Parallel data augmentations were done in an analogous way. For the raw-token-pair augmentation, we add in this task with a 5% weight and shrink the other weights accordingly. We use a task-specific token for each of the tasks the models may see, namely translation, MASS, GlowupMono, GlowupParallel, CodeswitchMono, and CodeswitchParallel.

6 Methods

In this paper, we divide our augmentation approaches into two classes: “codeswitching” approaches, which involve substituting source sentence words for their dictionary translations, and “GLOWUP,” which entails prepending dictionary translations of source words to source sentences. The main difference between these approaches is whether dictionary translations are substituted for source text (in the case of codeswitching) or added to the sentence (in the case of GLOWUP). As a third augmentation, we experiment with training on raw lexicon token pairs directly, treating them like any other parallel data.

The novelty of our contribution lies not so much in any one of our methods, but rather in (1) the application of these methods to unsupervised machine translation; (2) the number of methods we compare in controlled experiments; (3) the scale of our experiments, in terms of number of languages, data quantity, and model capacity; and (4) the application of these methods to “in the wild” web-crawled

data. While a variety of papers (e.g. Reid and Artetxe (2022); Yang et al. (2020)) have explored specific augmentations on particular language pairs, we believe our paper is the first to undertake a rigorous comparison of different augmentation strategies across hundreds of languages in a real-world setting.

6.1 Codeswitching

In our “codeswitching” augmentation strategy, words in the source sentence are swapped out for their dictionary translations to create mixed-language sentences. We experiment with this augmentation on both monolingual and parallel data. The details of this method are described below.

6.1.1 Codeswitched Autoencoding

Our multilingual codeswitching autoencoding (MCA) approach is similar to the “dictionary denoising” objective in Reid and Artetxe (2022). Let D represent a multilingual lexicon containing word or phrase translation pairs for many languages. Given a source sentence $x = (x_1, x_2, \dots, x_n)$ from monolingual corpus X_{mono} , we substitute each token in x for its dictionary translation with probability $p_{tr} = 0.4$. (More implementation details in Appendix Section H). Note that Reid and Artetxe (2022) also apply additional noise to x on top of codeswitching, along the lines of (m)BART (Lewis et al., 2020; Liu et al., 2020). For simplicity and so we can better examine the effects of lexicon information in isolation, we do not do this.

6.1.2 Codeswitching MT

Our codeswitching MT task is essentially the same approach as described in Section 6.1.1, except it applies to parallel rather than monolingual data. Given a source sentence x from parallel corpus X_{parallel} , we perform the identical procedure described in section Section 6.1.1 to obtain multilingual codeswitched sentence x' . We then train the model on the translation task using sentence pairs (x', y) , where (x, y) is a sentence pair in X_{parallel} . This method is effectively identical to the Random Aligned Substitution method proposed in Lin et al. (2020). As with MCA, we use $p_{tr} = 0.4$ and apply the augmentation on half the available parallel data.

6.2 Lexical prompting (GLOWUP)

The second class of lexical augmentations we experiment with is lexical prompting, which we

call GLOWUP (Guiding Lexical Outputs With Understandable Prompts). This method prepends $(src, transl)$ pairs to the beginning of a source sentence for some uniform random fraction of the words in that sentence that are in out blex. These hints can then be used to help the model guess the translation or the denoised sentence. The GLOWUP task has the advantage that it can be used at inference time, without retraining a model, and may be simpler to implement. However, it does result in longer and less balanced sequence lengths, which can pose problems for decoding.

6.2.1 MASS with monolingual GLOWUP

To apply GLOWUP to monolingual text, we simply first sample the lexical prompts, and after those translations are prepended to the source, we apply MASS in the standard way to mask random subspans. This may mean that the GLOWUP prompt itself is masked.

6.2.2 GLOWUP-MT

The extension of GLOWUP to parallel data (GLOWUP-MT), is effectively the same as the monolingual variant of the task, but without the MASS element. For a given sentence pair (x, y) in the training corpus, the prompting is performed on the source sentence x , essentially to give it hints about how to produce y . The model is then trained on the translation task using (x', y) , with the task token `<2glowup>` instead of `<2translation>`. Like GlowupMono, this can be applied in an inference-only way.

7 Experiments

7.1 Training regimes

In our experiments, we train models with various combinations of the augmentations outlined above, as well as a baseline (with no data augmentation of any kind) and a model where we simply provide word pairs from the lexicon as additional parallel data. The details of our training regimes are discussed below.

7.1.1 Baseline

We first train a baseline model with no data augmentation, using the monolingual and parallel data described in Sections 3.1 and 3.2, respectively. This model is comparable to the model trained in Bapna et al. (2022), but smaller, and without iterative back-translation; larger models with more data, are explored in Section 8.1.

7.1.2 Token-pair-only model

In addition to the baseline model, we also experiment with the extremely simple approach of providing raw word pairs from multilingual lexica to the model as additional parallel data. That is, given a dictionary entry s and its translation t , we provide the model with a “sentence” pair of the form $(\langle 2translation \rangle \langle 2lang \rangle \langle 2script \rangle s, t)$. We call this token-pair baseline GatiPanlexTokenPairs.

7.1.3 Single augmentation models

We also train models on each of the augmentations described in Section 6. As noted in that section, we only augment half the relevant data (monolingual, parallel, or both) before training each of these models, leaving the other half to be trained identically to the baseline (i.e. joint training on the MT task and MASS). The models with a single augmentation are named after their augmentation, viz. **CodeswitchMono**, **CodeswitchParallel**, **GlowupMono**, and **GlowupParallel**; those with two or more augmentations include them all in the name, viz. **CodeswitchMonoParallel**, **GlowupMonoParallel**, **CodeswitchMonoParallelGatiPanlex**, and **GlowupMonoParallelGatiPanlex**. We leave experimentation with a hybrid codeswitch-GLOWUP approach (e.g. CodeswitchMonoGlowupMono) for future work.

8 Results

We evaluate all our models on the FLORES-200 dataset (NLLBTeam et al., 2022; Goyal et al., 2021; Guzmán et al., 2019), which contains English-aligned parallel sentences for 200 languages. In Appendix section B we also report scores on the in-house GATONES eval set.

We use the following resourcedness classifications for our analysis:

1. High-Resource Languages (HRLs): $> 2B$ total training tokens in parallel data
2. Medium-Resource Languages (MRLs): 360M to 2B training tokens in parallel data
3. Low-Resource Languages (LRLs): 1 to 360M training tokens in parallel data
4. Unsupervised Languages (URLs): no parallel data

The results relative to the baseline, in ΔCHRF , are summarized in Tables 1 (en $\rightarrow$ xx) and 2 (xx $\rightarrow$ en). A few trends jump out. Firstly, all models trained with only monolingual data augmentations see consistent performance gains over

Model	μ	HRL	MRL	LRL	URL	LRL_{GAT}	URL_{GAT}
Baseline	39.7	50.5	46.6	34.4	28.7	35.4	26.6
GatiPanlexTokenPairs	+0.4	-0.3	-0.1	+0.5	+1.4	+1.8	+5.1
CodeswitchMono	+0.8	+0.8	+0.9	+0.8	+0.4	+1.9	+4.8
CodeswitchParallel	-1.4	-1.9	-1.7	-1.2	-0.9	+0.2	+2.1
CodeswitchMonoParallel	-0.2	-1.0	-0.7	+0.0	+1.1	+1.5	+5.2
CodeswitchMonoGatiPanlex	+1.0	+0.4	+0.6	+1.3	+1.5	+2.8	+7.0
GlowupMono	+1.1	+1.5	+1.4	+1.0	+0.5	+2.2	+5.6
GlowupParallel	-1.1	-2.0	-1.8	-1.1	+0.4	+0.8	+3.0
GlowupMonoParallel	+0.3	+0.1	+0.2	+0.1	+1.0	+1.3	+3.4
GlowupMonoGatiPanlex	+1.2	+1.3	+1.2	+1.4	+0.8	+2.5	+5.6

Table 1: en→xx performance on the FLORES-200 test set, measured in Δ CHRF over the baseline. Gains are particularly strong on the languages with GATITOS (last columns), reaching +7 CHRF for unsupervised language pairs.

Model	μ	HRL	MRL	LRL	URL	LRL_{GAT}	URL_{GAT}
Baseline	47.2	57.2	52.1	43.6	37.0	49.1	36.1
GatiPanlexTokenPairs	-0.0	-0.3	-0.3	+0.1	+0.5	+1.0	+2.4
CodeswitchMono	+0.2	+0.5	+0.4	+0.1	-0.1	+0.8	+1.5
CodeswitchParallel	-1.2	-1.6	-1.5	-1.2	-0.3	+0.0	+1.2
CodeswitchMonoParallel	-0.8	-0.7	-0.9	-0.8	-0.7	-0.1	+1.1
CodeswitchMonoGatiPanlex	+0.1	+0.3	+0.1	+0.3	-0.1	+0.9	+2.9
GlowupMono	+0.7	+1.2	+1.1	+0.6	+0.0	+1.5	+2.0
GlowupParallel	-1.3	-1.7	-1.6	-1.2	-0.6	-0.1	+1.2
GlowupMonoParallel	+0.1	+0.0	-0.0	-0.0	+0.5	+0.6	+1.7
GlowupMonoGatiPanlex	+0.6	+1.0	+0.8	+0.6	-0.1	+1.3	+1.8

Table 2: xx→en performance on the FLORES-200 test set, measured in Δ CHRF over the baseline, showing a weaker version of the same trends from en→xx.

the baseline. Conversely, models with only parallel data augmentations show performance degradations. Models mixing monolingual and parallel data augmentations fare in-between those poles.

These general trends are the same between en→xx and xx→en directions, though the gains are generally lower in the xx→en direction. As noted in Section 4, this is expected, and this direction is less of a priority for translation improvements. Results on GATONES, in Appendix B, show the same trends, though the performance gains tend to be larger for all augmentations, with CodeSwitch-MonoGatiPanlex gaining +2.3 CHRF for URLs.

Despite these trends seeming robust across models, the effect sizes are relatively small, maxing out at about +1.5 average Δ CHRF gain. However, the picture changes dramatically when only looking at the subset of languages that has the higher-quality GATITOS training lexica. For these 26 languages, every augmentation, even the parallel ones, have large performance gains. The win-

ning augmentations for URLs remain CodeSwitch-MonoGatiPanlex and GlowupMonoGatiPanlex, the former having an average gain of +7.0CHRF on FLORES-200 and +8.0CHRF on GATONES, and the latter having +5.6CHRF on FLORES-200 and +9.5CHRF on GATONES.

Finally, although it was not the first place model in any category, the GatiPanlexTokenPairs has large gains in all directions over baseline, and only falls short of the more complex augmentations by a small margin. Furthermore, when used in conjunction with either Glowup or Codeswitch, it further improves performance across all categories. This may be one of the most useful long-term findings of this study: raw token pairs perform roughly on par with all the other fancier augmentations!

8.1 Scaling up: bigger models, more data

Sometimes, results on smaller models do not transfer to larger models. To this end, we train larger transformer models with 1.6B parameters using

10× the parallel data and 10× the high-resource monolingual data.

We trained three large models: a baseline, a token-pair model, and a token-pairs + Codeswitch-Mono model. The CHRF scores for these models can be seen in Table 3. There are a number of obvious differences between these results and the results on the smaller (475M parameter) models trained with $\approx \frac{1}{10}$ th the data. First, the positive impact of the data augmentations is smaller in all categories, indicating that the gains previously seen from augmentations are partially washed out in the larger-data, larger model regime. On FLORES-200 en→xx, both models are very close to baseline—within the realm of noise. On GATONES en→xx, they see consistent small gains of around +0.5 CHRF, much smaller than previously. For both eval sets in the xx→en direction, there are consistent small losses.

Is this the Bitter Lesson (Sutton, 2019) getting us again? Perhaps—but the picture is less bleak than it first appears. When we look at the subset of the languages where we have a higher assurance that the bilingual lexica are higher-quality—namely, those that use GATITOS bilingual lexica—we still see consistent wins. For these 26 languages, all models see consistent gains, and the gains are biggest on unsupervised languages.

Overall, the takeaway from these experiments is that one has to ensure that the data is of high quality when applying lexical data augmentation at such a large scale. While we saw substantial improvements for many languages, these were balanced out by losses for other languages (especially those with only Panlex, but not GATITOS, data).

8.2 Glowup Decoding

In principle, one of the advantages of the Glowup-Parallel approach is that lexica can be used at inference time. Therefore, we experimented with decoding the eval sets not with the translation task ID, but with the Glowup task ID, along with the relevant lookups from the lexica. Unfortunately, these decodes failed impressively, with performance degrading the more prompts that were included. Model decodes often had long sequences of control tokens. Further work should not disregard this direction; indeed, a variant of this likely has particular promise in the world of foundation models. The current approach likely just needs some tweaks to eliminate this sort of out-of-domain

decoding errors we were seeing, but we leave an investigation of this hypothesis for future work.

8.3 Oracles: trusted parallel text

Parallel text is much more costly to produce than bilingual lexica, but also contains many more useful signals, including examples of word usage in context. But how much more helpful is it, really, than bilingual lexica? The answer seems to be “much more helpful”.

To measure this, we trained a model with a mixture of thirteen public parallel datasources (Appendix A) covering our lowest-resource languages. These are high-quality, trusted datasets, prepared by community members – a very different resource than the web-mined parallel data that the model is otherwise trained on.

Table 4 reports on the 24 GATITOS languages, comparing four models: the standard baseline and GatiPanlexTokenPairs model, as well as the “Parallel” model (which adds the external parallel translation task with a 5% weight) and the “Parallel + GatiPanlexTokenPairs” model, which uses both the token-pairs and the parallel data, with a combined weight of 5%. On both eval sets, we see that using Bilex yields a gain of around +3.5 CHRF, but using true parallel yields a much larger gain of about +10 CHRF. Using the bilingual lexica on top of the parallel data yields a further gain of about +0.5 CHRF, demonstrating that, though many of the gains have been washed out by the true parallel data, there are still modest gains to be had from bilex training. Full results are in Appendix Table 9.

8.4 How many token pairs do I really need?

We examine the relationship between the number of lexical token pairs provided during training and CHRF. First, we perform regressions using ΔCHRF over baseline as the outcome variable and three predictor variables: (1) number of Panlex entries, (2) number of GATITOS entries, and (3) number of monolingual sentences. We include monolingual sentences in the regression to control for it as a confound. To eliminate parallel data quantity as a confound, we only perform this analysis on URLs.

As expected, both the number of Panlex word pairs for a given language and the number of GATITOS word pairs have a positive β coefficient. However, note that the β for GATITOS is $\approx 3\times$ larger than β for Panlex on FLORES-200, and $\approx 39\times$ larger for GATONES. We conclude that GATITOS is

	HRL	MRL	LRL	URL	LRL _{GAT}	URL _{GAT}
FLORES-200 en→xx						
BaselineBig	56.1	51.2	38.5	35.5	24.4	36.0
GatiPanlexTokenPairsBig	-0.5	-0.3	+0.1	-0.3	+1.2	+2.2
CodeswitchMonoGatiPanlexBig	+0.1	+0.1	+0.3	-0.3	+0.9	+3.4
GATONES en→xx						
BaselineBig	-	34.1	26.9	25.5	23.9	27.1
GatiPanlexTokenPairsBig	-	+0.2	+0.6	+0.7	+1.6	+3.5
CodeswitchMonoGatiPanlexBig	-	+0.4	+0.7	+0.2	+1.4	+4.2

Table 3: Average CHRF scores by resource category for the larger models, reported in delta relative to baseline. These models are trained with 10x the data and 3x the parameters. The gains as a whole are washed out somewhat, but for those low-resource and unsupervised language pairs with the more trusted GATITOS data (last two columns), there is still a noticeable gain.

	Baseline	GatiPanlexTokenPairs	Parallel	Parallel + GatiPanlex
FLORES-200	21.1	24.4	33.6	34.2
GATONES	20.2	23.9	29.5	30.1

Table 4: Comparing en→xx improvements from token-pair data to the oracle: training on trusted parallel data. Parallel data is much more effective than GATITOS alone, but the combination of the two is the most effective overall.

more efficient for improving MT than Panlex, probably due to higher quality. Full regression results are given in Appendix Table 6.

The most practical question we can seek to answer is, **If I can spend \$X on translating tokens, how much quality increase can I expect?** To investigate this “bang for buck” question in a more controlled way, we observe the effects of the GATITOS dataset in isolation, without Panlex. We train a “GatiTokenPairs” model, which is identical to the “GatiPanlexTokenPairs” model, except the token-pair task has only GATITOS data. Thus, this tells us specifically what gains we can expect if we are to get 4,500 tokens’ worth per language.

The result of this experiment is a gain of +4.9 CHRF on FLORES-200 and +5.2 CHRF on GATONES, respectively, for en→xx URLs; the improvement for languages with some parallel data is reduced but significant, at +1.6/2.3 CHRF resp. Full results reported in Appendix Table 10.

9 Conclusions

In this paper we explore the ways that that augmenting training data with bilingual lexicon information can improve the performance of machine translation models on low-resource and unsupervised languages, and open-source the GATITOS dataset, which leads to average gains of about +7 CHRF on unsupervised languages. We perform extensive experimentation with three main types of lexical augmentation: codeswitching, lexical

prompting, and raw token-pair training. The results show that applying any of these augmentations to monolingual data yields substantial improvements, and that they can be combined for greater effect. The leader (by a small margin) is the combination of CodeswitchMono and raw token-pair training. These results hold when scaling up model and data size, but in the settings with more data and larger models, the quality of the bilingual lexica plays a relatively bigger role, and augmentation with the noisier Panlex begins to lag in quality behind the much smaller, yet higher-quality, GATITOS dataset.

Future work will likely want to focus on prompting foundation models with bilingual lexica. Large Language Models show promise on machine translation for high-resource languages (Jiao et al., 2023), but their capabilities on low-resource languages have yet to be thoroughly explored. Additionally, a more thorough investigation of the trade-off between cost and quality for tiny datasets can be explored: with a fixed budget of time or money, should one spend their time translating text, making monolingual text, or making bilingual lexica?

10 Limitations

There are several limitations with the present work. For one, we rely on the automated metric CHRF, which is less reliable than human ratings. Similarly, though we perform some qualitative evaluation of error types (Appendix G), a detailed human eval-

uation might reveal the precise ways in which our models are failing.

A second limitation is that, while we open-source GATITOS, the base training data from our models is not opensourced (to protect copyright), and thus our experiments are not replicable.

Finally, we leave open two important questions: 1) how much data do I need translated before I can reach X quality? and 2) if I can only translate X tokens’ worth of data, what is the best way to select those X tokens? The present work only partially answers the first question, and does not address the second at all.

11 Ethics Statement

Improving the state of technology for under-served communities is usually considered a positive contribution. There are various nuances to this, however, including questions of consent of and involvement of the affected community, “helicopter NLP,” and data sovereignty. By open-sourcing GATITOS to be used by all affected communities, we hope to respect their data sovereignty and not keep this resource from them. Throughout this project we have also sought advice from community members, which we hope will alleviate the other concerns.

References

- Antonov, Alexander. 2022. Chuvash-Russian parallel corpus. https://github.com/AlAntonov/chv_corpus.
- Roe Aharoni, Melvin Johnson, and Orhan Firat. 2019. **Massively multilingual neural machine translation**. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 3874–3884, Minneapolis, Minnesota. Association for Computational Linguistics.
- Benjamin Akera, Jonathan Mukiibi, Lydia Sanyu Nagayi, Claire Babirye, Isaac Owomugisha, Solomon Nsumba, Joyce Nakatumba-Nabende, Engineer Bainomugisha, Ernest Mwebaze, and John Quinn. 2022. **Machine translation for african languages: Community creation of datasets and models in uganda**. In *3rd Workshop on African Natural Language Processing*.
- Andersen, Jógvan. 2021. Sprotin Translations. <https://github.com/Sprotin/translations>.
- Rosana Ardila, Megan Branson, Kelly Davis, Michael Kohler, Josh Meyer, Michael Henretty, Reuben Morais, Lindsay Saunders, Francis Tyers, and Gregor Weber. 2020. **Common voice: A massively-multilingual speech corpus**. In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pages 4218–4222, Marseille, France. European Language Resources Association.
- Mikel Artetxe, Gorka Labaka, and Eneko Agirre. 2019. **An effective approach to unsupervised machine translation**. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 194–203, Florence, Italy. Association for Computational Linguistics.
- Mikel Artetxe, Gorka Labaka, Eneko Agirre, and Kyunghyun Cho. 2017. **Unsupervised neural machine translation**. *CoRR*, abs/1710.11041.
- Dzmitry Bahdanau, Kyunghyun Cho, and Yoshua Bengio. 2015. **Neural machine translation by jointly learning to align and translate**. In *3rd International Conference on Learning Representations, ICLR 2015, San Diego, CA, USA, May 7-9, 2015, Conference Track Proceedings*.
- Marta Bañón, Pinzhen Chen, Barry Haddow, Kenneth Heafield, Hieu Hoang, Miquel Esplà-Gomis, Mikel L. Forcada, Amir Kamran, Faheem Kirefu, Philipp Koehn, Sergio Ortiz Rojas, Leopoldo Pla Sempere, Gema Ramírez-Sánchez, Elsa Sarriás, Marek Strelec, Brian Thompson, William Waites, Dion Wiggins, and Jaume Zaragoza. 2020. **ParaCrawl: Web-scale acquisition of parallel corpora**. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4555–4567, Online. Association for Computational Linguistics.
- Ankur Bapna, Isaac Caswell, Julia Kreutzer, Orhan Firat, Daan van Esch, Aditya Siddhant, Mengmeng Niu, Pallavi Baljekar, Xavier Garcia, Wolfgang Macherey, Theresa Breiner, Vera Axelrod, Jason Riesa, Yuan Cao, Mia Xu Chen, Klaus Macherey, Maxim Krikun, Pidong Wang, Alexander Gutkin, Apurva Shah, Yanping Huang, Zhifeng Chen, Yonghui Wu, and Macduff Hughes. 2022. **Building Machine Translation Systems for the Next Thousand Languages**. *arXiv e-prints*, page arXiv:2205.03983.
- Christos Baziotis, Barry Haddow, and Alexandra Birch. 2020. **Language model prior for low-resource neural machine translation**. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 7622–7634, Online. Association for Computational Linguistics.
- Isaac Caswell, Theresa Breiner, Daan van Esch, and Ankur Bapna. 2020. **Language id in the wild: Unexpected challenges on the path to a thousand-language web text corpus**.
- Isaac Caswell, Ciprian Chelba, and David Grangier. 2019. **Tagged back-translation**. In *Proceedings of the Fourth Conference on Machine Translation (Volume 1: Research Papers)*, pages 53–63, Florence, Italy. Association for Computational Linguistics.
- Aditi Chaudhary, Karthik Raman, Krishna Srinivasan, and Jiecao Chen. 2020. **DICT-MLM: Improved Multilingual Pre-Training using Bilingual Dictionaries**. *arXiv e-prints*, page arXiv:2010.12566.

- Yong Cheng, Wei Wang, Lu Jiang, and Wolfgang Macherey. 2021. [Self-supervised and supervised joint training for resource-rich machine translation](#). In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 1825–1835. PMLR.
- Luis Chiruzzo, Santiago Góngora, Aldo Alvarez, Gustavo Giménez-Lugo, Marvin Agüero-Torales, and Yliana Rodríguez. 2022. [Jojajovai: A parallel Guarani-Spanish corpus for MT benchmarking](#). In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 2098–2107, Marseille, France. European Language Resources Association.
- Alexis Conneau, Guillaume Lample, Marc’Aurelio Ranzato, Ludovic Denoyer, and Hervé Jégou. 2017. Word translation without parallel data. *arXiv preprint arXiv:1710.04087*.
- Chris Chinenye Emezue and Femi Pancrace Bonaventure Dossou. 2020. [FFR v1.1: Fon-French neural machine translation](#). In *Proceedings of the The Fourth Widening Natural Language Processing Workshop*, pages 83–87, Seattle, USA. Association for Computational Linguistics.
- Miquel Esplà-Gomis. 2009. [Bitextor: a free/open-source software to harvest translation memories from multilingual websites](#). In *Beyond Translation Memories: New Tools for Translators Workshop*, Ottawa, Canada.
- Angela Fan, Shruti Bhosale, Holger Schwenk, Zhiyi Ma, Ahmed El-Kishky, Siddharth Goyal, Man-deep Baines, Onur Celebi, Guillaume Wenzek, Vishrav Chaudhary, Naman Goyal, Tom Birch, Vitaliy Liptchinsky, Sergey Edunov, Edouard Grave, Michael Auli, and Armand Joulin. 2022. Beyond english-centric multilingual machine translation. *J. Mach. Learn. Res.*, 22(1).
- Isaac Feldman and Rolando Coto-Solano. 2020. [Neural machine translation models with back-translation for the extremely low-resource indigenous language Bribri](#). In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 3965–3976, Barcelona, Spain (Online). International Committee on Computational Linguistics.
- Orhan Firat, Kyunghyun Cho, and Yoshua Bengio. 2016. [Multi-way, multilingual neural machine translation with a shared attention mechanism](#). In *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 866–875, San Diego, California. Association for Computational Linguistics.
- Markus Freitag, Ricardo Rei, Nitika Mathur, Chi kiu Lo, Craig Stewart, Eleftherios Avramidis, George Foster, Alon Lavie, and Andr’ e F. T. Martins. 2022a. [Results of wmt22 metrics shared task: Stop using bleu – neural metrics are better and more robust](#).
- Markus Freitag, David Vilar, David Grangier, Colin Cherry, and George Foster. 2022b. [A natural diet: Towards improving naturalness of machine translation output](#). In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 3340–3353, Dublin, Ireland. Association for Computational Linguistics.
- Marjan Ghazvininejad, Hila Gonen, and Luke Zettlemoyer. 2023. [Dictionary-based phrase-level prompting of large language models for machine translation](#).
- Naman Goyal, Cynthia Gao, Vishrav Chaudhary, Peng-Jen Chen, Guillaume Wenzek, Da Ju, Sanjana Krishnan, Marc’Aurelio Ranzato, Francisco Guzmán, and Angela Fan. 2021. The flores-101 evaluation benchmark for low-resource and multilingual machine translation.
- Caglar Gulcehre, Orhan Firat, Kelvin Xu, Kyunghyun Cho, and Yoshua Bengio. 2017. [On integrating a language model into neural machine translation](#). *Computer Speech & Language*, 45:137–148.
- Francisco Guzmán, Peng-Jen Chen, Myle Ott, Juan Pino, Guillaume Lample, Philipp Koehn, Vishrav Chaudhary, and Marc’Aurelio Ranzato. 2019. Two new evaluation datasets for low-resource machine translation: Nepali-english and sinhala-english.
- Asmelash Hadgu, Gebrekirstos Gebremeskel, and Abel Aregawi. 2022. HornMT. <https://github.com/asmelashteka/HornMT>. Accessed: 2023-03-24.
- Wenxiang Jiao, Wenxuan Wang, Jen-tse Huang, Xing Wang, and Zhaopeng Tu. 2023. [Is chatgpt a good translator? a preliminary study](#).
- Melvin Johnson, Mike Schuster, Quoc V. Le, Maxim Krikun, Yonghui Wu, Zhifeng Chen, Nikhil Thorat, Fernanda Viégas, Martin Wattenberg, Greg Corrado, Macduff Hughes, and Jeffrey Dean. 2017. [Google’s multilingual neural machine translation system: Enabling zero-shot translation](#). *Transactions of the Association for Computational Linguistics*, 5:339–351.
- David Kamholz, Jonathan Pool, and Susan Colowick. 2014. [PanLex: Building a resource for panlingual lexical translation](#). In *Proceedings of the Ninth International Conference on Language Resources and Evaluation (LREC’14)*, pages 3145–3150, Reykjavik, Iceland. European Language Resources Association (ELRA).
- Jyotsana Khatri, Rudra Murthy, Tamali Banerjee, and Pushpak Bhattacharyya. 2021. [Simple measures of bridging lexical divergence help unsupervised neural machine translation for low-resource languages](#).
- Tom Kocmi, Christian Federmann, Roman Grundkiewicz, Marcin Junczys-Dowmunt, Hitokazu Matsushita, and Arul Menezes. 2021. [To ship or not to ship: An extensive evaluation of automatic metrics for machine translation](#). In *Proceedings of the Sixth Conference on Machine Translation*, pages 478–494, Online. Association for Computational Linguistics.

- Julia Kreutzer, Isaac Caswell, Lisa Wang, Ahsan Wahab, Daan van Esch, Nasanbayar Ulzii-Orshikh, Allahsera Tapo, Nishant Subramani, Artem Sokolov, Claytone Sikasote, Monang Setyawan, Supheakmongkol Sarin, Sokhar Samb, Benoît Sagot, Clara Rivera, Annette Rios, Isabel Papadimitriou, Salomey Osei, Pedro Ortiz Suarez, Iroko Orife, Kelechi Ogueji, Andre Niyongabo Rubungo, Toan Q. Nguyen, Mathias Müller, André Müller, Shamsuddeen Hassan Muhammad, Nanda Muhammad, Ayanda Mnyakeni, Jamshidbek Mirzakhalov, Tapiwanashe Matangira, Colin Leong, Nze Lawson, Sneha Kudugunta, Yacine Jernite, Mathias Jenny, Orhan Firat, Bonaventure F. P. Dossou, Sakhile Dlamini, Nisansa de Silva, Sakine Çabuk Ballı, Stella Biderman, Alessia Battisti, Ahmed Baruwa, Ankur Bapna, Pallavi Baljekar, Israel Abebe Azime, Ayodele Awokoya, Duygu Ataman, Orevaoghene Ahia, Oghenefego Ahia, Sweta Agrawal, and Mofetoluwa Adeyemi. 2022. [Quality at a glance: An audit of web-crawled multilingual datasets](#). *Transactions of the Association for Computational Linguistics*, 10:50–72.
- Nalin Kumar, Deepak Kumar, and Subhankar Mishra. 2022. [Dict-nmt: Bilingual dictionary based nmt for extremely low resource languages](#).
- Garry Kuwanto, Afra Feyza Akyürek, Isidora Chara Tourni, Siyang Li, and Derry Wijaya. 2021. [Low-resource machine translation for low-resource languages: Leveraging comparable data, code-switching and compute resources](#). *CoRR*, abs/2103.13272.
- Guillaume Lample, Alexis Conneau, Ludovic Denoyer, and Marc’Aurelio Ranzato. 2017. Unsupervised machine translation using monolingual corpora only. *arXiv preprint arXiv:1711.00043*.
- Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Veselin Stoyanov, and Luke Zettlemoyer. 2020. [BART: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 7871–7880, Online. Association for Computational Linguistics.
- Yusen Lin, Jiayong Lin, Shuaicheng Zhang, and Haoying Dai. 2021. [Bilingual dictionary-based language model pretraining for neural machine translation](#). *CoRR*, abs/2103.07040.
- Zehui Lin, Xiao Pan, Mingxuan Wang, Xipeng Qiu, Jiangtao Feng, Hao Zhou, and Lei Li. 2020. [Pre-training multilingual neural machine translation by leveraging alignment information](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2649–2663, Online. Association for Computational Linguistics.
- Yinhan Liu, Jiatao Gu, Naman Goyal, Xian Li, Sergey Edunov, Marjan Ghazvininejad, Mike Lewis, and Luke Zettlemoyer. 2020. [Multilingual denoising pre-training for neural machine translation](#). *Transactions of the Association for Computational Linguistics*, 8:726–742.
- Zihan Liu, Genta Indra Winata, and Pascale Fung. 2021. [Continual mixed-language pre-training for extremely low-resource neural machine translation](#). In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pages 2706–2718, Online. Association for Computational Linguistics.
- Manuel Mager, Arturo Oncevay, Abteen Ebrahimi, John Ortega, Annette Rios, Angela Fan, Ximena Gutierrez-Vasques, Luis Chiruzzo, Gustavo Giménez-Lugo, Ricardo Ramos, Ivan Vladimir Meza Ruiz, Rolando Coto-Solano, Alexis Palmer, Elisabeth Mager-Hois, Vishrav Chaudhary, Graham Neubig, Ngoc Thang Vu, and Katharina Kann. 2021. [Findings of the AmericasNLP 2021 shared task on open machine translation for indigenous languages of the Americas](#). In *Proceedings of the First Workshop on Natural Language Processing for Indigenous Languages of the Americas*, pages 202–217, Online. Association for Computational Linguistics.
- Ayush Maheshwari, Piyush Sharma, Preethi Jyothi, and Ganesh Ramakrishnan. 2022. [Dictdis: Dictionary constrained disambiguation for improved nmt](#).
- Christopher Malon. 2021. [Overcoming poor word embeddings with word definitions](#). *CoRR*, abs/2103.03842.
- Elise Michon, Josep Crego, and Jean Senellart. 2020. [Integrating domain terminology into neural machine translation](#). In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 3925–3937, Barcelona, Spain (Online). International Committee on Computational Linguistics.
- Jonathan Mukiibi, Babirye Claire, and Nakatumba-Nabende Joyce. 2021. [The Makerere MT Corpus: English to Luganda parallel corpus](#).
- Sreyashi Nag, Mihir Kale, Varun Lakshminarasimhan, and Swapnil Singhavi. 2020. [Incorporating bilingual dictionaries for low resource semi-supervised neural machine translation](#). *CoRR*, abs/2004.02071.
- Jan Niehues. 2021. [Continuous learning in neural machine translation using bilingual dictionaries](#). *CoRR*, abs/2102.06558.
- NLLBTeam, Marta R. Costa-jussà, James Cross, Onur Çelebi, Maha Elbayad, Kenneth Heafield, Kevin Hefernan, Elahe Kalbassi, Janice Lam, Daniel Licht, Jean Maillard, Anna Sun, Skyler Wang, Guillaume Wenzek, Al Youngblood, Bapi Akula, Loic Barrault, Gabriel Mejia Gonzalez, Prangthip Hansanti, John Hoffman, Semarley Jarrett, Kaushik Ram Sadagopan, Dirk Rowe, Shannon Spruit, Chau Tran, Pierre Andrews, Necip Fazil Ayan, Shruti Bhosale, Sergey Edunov, Angela Fan, Cynthia Gao, Vedanuj Goswami, Francisco Guzmán, Philipp

- Koehn, Alexandre Mourachko, Christophe Ropers, Safiyyah Saleem, Holger Schwenk, and Jeff Wang. 2022. No language left behind: Scaling human-centered machine translation.
- Xiao Pan, Mingxuan Wang, Liwei Wu, and Lei Li. 2021. [Contrastive learning for many-to-many multilingual neural machine translation](#). In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 244–258, Online. Association for Computational Linguistics.
- Matt Post. 2018. [A call for clarity in reporting BLEU scores](#). In *Proceedings of the Third Conference on Machine Translation: Research Papers*, pages 186–191, Brussels, Belgium. Association for Computational Linguistics.
- Libo Qin, Minheng Ni, Yue Zhang, and Wanxiang Che. 2020. [Cosda-ml: Multi-lingual code-switching data augmentation for zero-shot cross-lingual nlp](#). In *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence, IJCAI-20*, pages 3853–3860. International Joint Conferences on Artificial Intelligence Organization. Main track.
- Machel Reid and Mikel Artetxe. 2022. [PARADISE: Exploiting parallel data for multilingual sequence-to-sequence pretraining](#). In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 800–810, Seattle, United States. Association for Computational Linguistics.
- Philip Resnik and Noah A. Smith. 2003. [The web as a parallel corpus](#). *Computational Linguistics*, 29(3):349–380.
- Thibault Sellam, Dipanjan Das, and Ankur P. Parikh. 2020. [BLEURT: learning robust metrics for text generation](#). *CoRR*, abs/2004.04696.
- Rico Sennrich, Barry Haddow, and Alexandra Birch. 2016. [Improving neural machine translation models with monolingual data](#). In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 86–96, Berlin, Germany. Association for Computational Linguistics.
- Iskander Shakirov and Aigiz Kunafin. 2023. Bashkir-russian parallel corpora.
- Aditya Siddhant, Ankur Bapna, Yuan Cao, Orhan Firat, Mia Chen, Sneha Kudugunta, Naveen Arivazhagan, and Yonghui Wu. 2020. [Leveraging monolingual data with self-supervision for multilingual neural machine translation](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 2827–2835, Online. Association for Computational Linguistics.
- Aditya Siddhant, Ankur Bapna, Orhan Firat, Yuan Cao, Mia Xu Chen, Isaac Caswell, and Xavier Garcia. 2022. [Towards the next 1000 languages in multilingual machine translation: Exploring the synergy between supervised and self-supervised learning](#). *CoRR*, abs/2201.03110.
- Kai Song, Yue Zhang, Heng Yu, Weihua Luo, Kun Wang, and Min Zhang. 2019a. [Code-switching for enhancing NMT with pre-specified translation](#). In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 449–459, Minneapolis, Minnesota. Association for Computational Linguistics.
- Kaitao Song, Xu Tan, Tao Qin, Jianfeng Lu, and Tie-Yan Liu. 2019b. [MASS: Masked sequence to sequence pre-training for language generation](#). In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 5926–5936. PMLR.
- Raymond Hendy Susanto, Shamil Chollampatt, and Lil-ing Tan. 2020. [Lexically constrained neural machine translation with levenshtein transformer](#). *CoRR*, abs/2004.12681.
- Ilya Sutskever, Oriol Vinyals, and Quoc V Le. 2014. [Sequence to sequence learning with neural networks](#). In *Advances in Neural Information Processing Systems*, volume 27. Curran Associates, Inc.
- Richard Sutton. 2019. [The bitter lesson](#).
- Jörg Tiedemann. 2020. [The tatoeba translation challenge - realistic data sets for low resource and multilingual MT](#). *CoRR*, abs/2010.06354.
- Tlisha, Nart. 2022. Multilingual Parallel Corpus. <https://github.com/danielinux77/Multilingual-Parallel-Corpus>. Accessed: 2023-03-24.
- Jakob Uszkoreit, Jay Ponte, Ashok Popat, and Moshe Dubiner. 2010. [Large scale parallel document mining for machine translation](#). In *Proceedings of the 23rd International Conference on Computational Linguistics (Coling 2010)*, pages 1101–1109, Beijing, China. Coling 2010 Organizing Committee.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. [Attention is all you need](#). In *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc.
- Xinyi Wang, Sebastian Ruder, and Graham Neubig. 2022. [Expanding pretrained models to thousands more languages via lexicon-based adaptation](#). In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 863–877, Dublin, Ireland. Association for Computational Linguistics.

- Barack Wanjawa, Lilian Wanzare, Florence Indede, Owen McOnyango, Edward Ombui, and Lawrence Muchemi. 2022. [Kencorpus: A kenyan language corpus of swahili, dholuo and luhya for natural language processing tasks](#).
- Yonghui Wu, Mike Schuster, Zhifeng Chen, Quoc V. Le, Mohammad Norouzi, Wolfgang Macherey, Maxim Krikun, Yuan Cao, Qin Gao, Klaus Macherey, Jeff Klingner, Apurva Shah, Melvin Johnson, Xiaobing Liu, Łukasz Kaiser, Stephan Gouws, Yoshikiyo Kato, Taku Kudo, Hideto Kazawa, Keith Stevens, George Kurian, Nishant Patil, Wei Wang, Cliff Young, Jason Smith, Jason Riesa, Alex Rudnick, Oriol Vinyals, Greg Corrado, Macduff Hughes, and Jeffrey Dean. 2016. [Google’s neural machine translation system: Bridging the gap between human and machine translation](#). *CoRR*, abs/1609.08144.
- Mengzhou Xia, Xiang Kong, Antonios Anastasopoulos, and Graham Neubig. 2019. [Generalized data augmentation for low-resource translation](#). *CoRR*, abs/1906.03785.
- Jian Yang, Yuwei Yin, Shuming Ma, Haoyang Huang, Dongdong Zhang, Zhoujun Li, and Furu Wei. 2021. [Multilingual agreement for multilingual neural machine translation](#). In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 2: Short Papers)*, pages 233–239, Online. Association for Computational Linguistics.
- Zhen Yang, Bojie Hu, Ambyera Han, Shen Huang, and Qi Ju. 2020. [CSP:code-switching pre-training for neural machine translation](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2624–2636, Online. Association for Computational Linguistics.
- Wenhao Yu, Chenguang Zhu, Yuwei Fang, Donghan Yu, Shuohang Wang, Yichong Xu, Michael Zeng, and Meng Jiang. 2021. [Dict-bert: Enhancing language model pre-training with dictionary](#). *CoRR*, abs/2110.06490.
- Xing Jie Zhong and David Chiang. 2020. [Look it up: Bilingual and monolingual dictionaries improve neural machine translation](#). *CoRR*, abs/2010.05997.

GlowupMono w/ masking (multilingual lexically-prompted MASS)

Source <2glowup> <2en> <2Latn> <src> we <tgt> nous <src> see <tgt> ka
ye <2en> <MASK> Saturday, we drove <MASK> <MASK> see <MASK> countryside.
Target Last Saturday, we drove out to see the countryside.

GlowupParallel (target-language lexically-prompted translation)

Source <2glowup> <2fr> <2Latn> <src> came <tgt> venu <src> dramatic
<tgt> dramatique <src> fashion <tgt> façon <2fr> The performance came to
a close in dramatic fashion.
Target La performance s'est conclue de façon dramatique.

Figure 1: Examples of the monolingual (top) and parallel (bottom) GLOWUP augmentation strategies. In both cases, random tokens in the source sentence are prepended to the source sentence, along with their translations in a random language from a multilingual lexicon. As in Figure 2, differential color coding is done to draw attention to dictionary translations in different languages.

CodeswitchMono (multilingual codeswitching autoencoding)

Source <2codeswitch> <2en> <2Latn> Dernier Saturday, noi yatwaye out to
हेर्नुहोस् the panpa.
Target Last Saturday, we drove out to see the countryside.

CodeswitchParallel (codeswitching translation)

Source <2codeswitch> <2fr> <2Latn> The (паксма came til a близкий in
dramatisk fashion.
Target La performance s'est conclue de façon dramatique.

Figure 2: Examples of the monolingual (top) and parallel (bottom) codeswitching augmentation strategies. In both cases, random tokens in the source sentence are replaced with their translations in a random language from a multilingual lexicon. Color coding is used to indicate which source words have been swapped for their dictionary translation. The different colors are used simply to point out the fact that the words are from codeswitched words in each sentence come from different languages.

A Public parallel data used

The datasets we used were HornMT (Hadgu et al., 2022), SALT (Sunbird AI Language Translation) dataset (Akeru et al., 2022), FFR: Fon-French Neural Machine Translation (Emezue and Dos-sou, 2020), Tatoeba (Tiedemann, 2020), The Makerere MT Corpus: English to Luganda parallel corpus (Mukiibi et al., 2021), Commonvoice (Ardila et al., 2020), Kencorpus: Kenyan Languages Corpus (Wanjawa et al., 2022), Chuvash-Russian parallel corpus (Antonov, Alexander, 2022), Abkhaz Corpus (Tlisha, Nart, 2022), Bashkir Corpus (Shakirov and Kunafin, 2023), Sprotin Faroese Corpus (Andersen, Jógván, 2021), Jojajovai Guarani-Spanish Parallel Corpus (Chiruzzo et al., 2022), and the NLLB Seed data (NLLBTeam et al.,

2022). We were not able to use AmericasNLP (Mager et al., 2021) because of the license.

Source	Khawvelah hian tawng chi hrang 5000 chuang a awma, heng zinga tawng 20 ai tam hi chuan tawng hmanctu maktaduai 50 aia tam a nei a ni.
Reference	The world has over 5,000 different languages , including more than twenty with 50 million or more speakers.
Baseline	There are 5000 houses in the city, 5000 houses in the city, and 5000 houses in the city.
Augment	There are 5000 languages in the world, 20 of which are 50 million languages .
Source	Maani isumaqarpunga inuunerma sinnerani pilluurtussanngorlunga.
Reference	Here I think I'll remain happy throughout my life.
Baseline	I believe I am the only one who has ever died.
Augment	I mean, I'm blessed with a lifetime of joy.
Source	нагахъ вы ж хъашто хилар в дІадахар къамелан ма расширяемости, дазар с нами хула uservoice
Reference	If you are interested in continuing the conversation around extensibility, connect with us via uservoice
Baseline	If you want to use the extension, see uservoice
Augment	If you want to learn more about extensibility, contact us at uservoice

Table 5: Examples of translations where our lexical data augmentation methods appear to have helped the model choose the correct vocabulary, in Mizo (lus), Kalaallisut (kl), and Chechen (ce). Words appearing in the lexical data are colored green.

	No. PanLex	No. GAT.	No. mono	Intercept	Adjusted R^2
Δ FLORES-200	6.55e-5	1.96e-4	-6.71e-7	1.93	0.40
Δ GATONES	6.59e-6	2.58e-4	-7.78e-8	2.91	0.23

Table 6: Linear regression results in the en $\rightarrow$ xx direction. Here, the dependent variable is Δ CHRF over the baseline, and the independent variables are: number of PanLex word pairs for a given language (No. PanLex), number of GATITOS word pairs (No. GATITOS), and number of monolingual sentences (No. mono).

B Results on GATONES

The main paper reports the scores on the more widely-used FLORES-200 eval set; this section reports on the other dataset that we evaluate our models on. This is an in-house eval set, which we call GATONES (Google AuTOMatic NTL Eval Set). The dataset contains 63 languages, most of which are unsupervised or low-resource. Here are some notes, by translation direction:

en $\rightarrow$ xx The evaluation results on GATONES for the en $\rightarrow$ xx direction are summarized in Table 7. The trends are mostly the same as what we saw for the FLORES-200 en $\rightarrow$ xx evaluation. Codeswitch-MonoGatiPanlex emerges, even more definitively than on FLORES-200, as the best model, winning all categories except MRLs.

xx $\rightarrow$ en The evaluation results for the xx $\rightarrow$ en direction are given in Table 8. The single most important takeaway from this part of the analysis is the same as it was for the FLORES-200 evaluation: the plain GatiPanlexTokenPairs model helps URLs the most in this direction, with a Δ CHRF of +1.1 over the baseline. Yet again, the improvements are smaller in this direction than for en $\rightarrow$ xx. The only other thing that stands out about this part of the evaluation is that the GlowupMono augmentation doesn't seem to be as helpful according to this test set as for the FLORES-200 set. Although GlowupMonoParallel and GlowupMonoGatiPanlex do reasonably well, their improvements are sig-

nificantly smaller than the improvement from using GatiPanlexTokenPairs alone, and the GlowupMono augmentation by itself actually results in losses on URLs. So taking the GATONES and FLORES-200 results together, it seems that adding raw token pairs as additional parallel data is the best way, out of the techniques we tried, to improve performance in the xx $\rightarrow$ en direction for very low-resource languages.

C Bilexica vs true parallel data

In the main paper in Section 8.3, we evaluate on the 24 GATITOS languages, comparing four models with different combinations of trusted parallel data and GATITOS bilingual lexica. The full results are given here in Table 9, on both FLORES-200 and GATONES, sorted from highest-resource to lowest-resource.

Model	μ	MRL	LRL	URL	LRL _{GAT}	URL _{GAT}
Baseline	21.4	28.7	22.9	19.1	17.1	20.0
GatiPanlexTokenPairs	+1.7	+0.6	+1.6	+1.8	+3.6	+8.4
CodeswitchMono	+1.4	+1.6	+0.8	+1.9	+3.2	+8.3
CodeswitchParallel	-0.5	-1.8	-0.9	-0.0	+0.9	+3.4
CodeswitchMonoParallel	+1.4	-0.1	+0.7	+2.2	+2.5	+6.1
CodeswitchMonoGatiPanlex	+2.2	+1.1	+2.1	+2.3	+3.9	+8.0
GlowupMono	+0.3	+1.0	+0.6	-0.2	+2.0	+7.6
GlowupParallel	-0.2	-1.6	-0.7	+0.6	+1.0	+3.0
GlowupMonoParallel	+0.2	-0.3	-0.7	+1.1	+2.1	+7.6
GlowupMonoGatiPanlex	+1.3	+1.4	+1.9	+0.7	+3.5	+9.5

Table 7: en→xx performance on the GATONES test set, measured in Δ CHRF over the baseline.

Model	μ	MRL	LRL	URL	LRL _{GAT}	URL _{GAT}
Baseline	29.4	43.2	30.2	27.0	24.7	22.1
GatiPanlexTokenPairs	+0.8	+0.0	+0.5	+1.1	+1.1	+3.2
CodeswitchMono	+0.1	+0.1	+0.0	+0.2	+0.8	+2.2
CodeswitchParallel	-0.4	-1.2	-0.9	+0.2	+0.3	+1.8
CodeswitchMonoParallel	-0.5	-0.8	-0.7	-0.2	+0.2	+1.6
CodeswitchMonoGatiPanlex	+0.5	+0.0	+0.5	+0.6	+1.1	+3.3
GlowupMono	-0.0	+0.9	+0.2	-0.4	+0.5	+1.7
GlowupParallel	-0.6	-1.5	-0.9	-0.3	+0.0	+1.3
GlowupMonoParallel	+0.0	-0.2	-0.2	+0.2	+0.2	+1.6
GlowupMonoGatiPanlex	+0.2	+0.8	+0.3	+0.1	+0.8	+3.3

Table 8: xx→en performance on the GATONES test set, measured in Δ CHRF over the baseline.

	mean	ts	nso	lg	ee	bho	bm	ff	gn	ti	om
en→xx Parallel toks	375169	2296670	606853	315905	275417	154023	148549	130953	112958	43511	42018
en→xx Bilex toks	16324	8015	4500	8298	6755	24665	24665	70854	4500	6484	4500
FLORES-200 en→xx											
Baseline	21.1	33.2	31.9	30.1	26.9	14.7	15	19.1	19.1	5.9	15.5
GatiPanlexTokenPairs	24.4	37.3	31.9	32.7	30.5	25.5	22.6	19.6	19.4	8.1	16.1
Parallel	33.6	45.6	48	39.4	28	40.6	30.7	25	39.2	15.7	24.1
Parallel + GatiPanlex	34.2	45.3	45.3	38.2	31.1	40.8	30.1	24.9	39.9	18.1	28.8
GATONES en→xx											
Baseline	20.2	32.1	28.3	27.6	23.5	15.8	13.5	23.9	15.6	6.9	14.6
GatiPanlexTokenPairs	23.9	37.3	28.7	31.3	28.1	26.2	22.5	23.5	16.1	9.3	15.6
Parallel	29.5	42.9	39.9	35.5	26.9	38.5	30.3	23.6	28.2	11.1	18
Parallel + GatiPanlexTokenPairs	30.1	42.6	37.9	35	29.6	38.7	30.4	22.6	30.1	13.1	20.8

Table 9: Comparing improvements from token-pair data to the oracle: training on trusted parallel data.

D Training only on GATITOS

The main paper (Section 8.4) describes the results of training models only on GATITOS and not on PanLex. Full results are reported here, in Table 10, reported in delta versus the baseline model for both FLORES-200 and GATONES in the en→xx direction. The improvement for unsupervised languages is around +5.0 CHRF for both eval sets; the improvement for languages with some parallel data is less but still noticeable, hovering around +2 CHRF. The largest improvement is in Goan Konkani (+11.0 CHRF), with Mizo, Ilocano, and Bambara close on its heels with gains of around +8 CHRF. Only Maithili, which has interesting properties as a close dialect of Hindustani, sees a loss on both eval sets. The gains are not obviously related to the total number of tokens per language.

As an aside, it is heartening that FLORES-200 and GATONES seem to agree very nicely, despite their different domains (Wikipedia versus web + question-answers).

TOTAL TOKENS			1.8M	2.1M	2.6M	3.6M	3.7M	5.4M	6.2M	14.7M	14.8M	16M	16.9M	26.1M
	μ_{lrl}	μ_{url}	ff	mni-Mtei	kri	doi	bm	ay	gom	bho	kl	ee	qu	ts
Δ FLORES-200	+1.6	+4.9	+0.1	-	-	-	+5.0	+0.8	-	+4.5	-	+2.9	-	+4.2
Δ GATONES	+2.3	+5.2	+0.0	+0.7	+5.1	+1.5	+7.6	-0.1	+11.0	+5.1	+4.8	+3.9	+1.7	+5.3
TOTAL TOKENS	26.4M	27M	28M	40M	41M	52M	52M	80M	115M	124M	157M	167M	204M	505M
	ak	mai	ln	lg	gn	nso	ilo	ti	om	sa	dv	lus	as	ckb
Δ FLORES-200	-	-1.2	+1.3	+1.6	-0.4	-0.5	+7.0	+0.9	-0.2	+0.2	-	+8.6	+3.0	+3.3
Δ GATONES	+2.0	-1.2	1.2	+3.0	+0.5	+0.2	+8.0	+1.7	+0.1	+0.4	+6.1	+9.3	+3.4	+2.7

Table 10: Improvements on languages with GATITOS data when training ONLY on GATITOS data, in the en→xx direction. Sorted by total training tokens (mono, parallel, and biled), in millions.

E Effects on distributionally similar noun mistranslation

Part of the motivation for using bilingual lexicons for unsupervised translation was to see whether we could repair the common error mode of mistranslating distributionally similar nouns. Bapna et al. (2022) note that this error mode is particularly common for two categories of nouns: animals and colors.

To measure improvement on this phenomenon, we define the *token hit-rate* as the following: for some set of desired tokens D , let R_D be the subset of the eval set such that each reference contains at least one token in D . The hit-rate is then the percentage of times in R_D that the model correctly generated one of the desired tokens in D . For instance, if the desired tokens are “kitten” and “puma”, R_D is the set of references containing one of these words, e.g. “The **kitten** lies” and “A **Puma** eats hot chip”. If the model produces “**kitten** lie on floor” and “**Crocodile** charge they phone” from the corresponding sources, it has a hit-rate of 50%, since it correctly got “kitten”, but not “puma”.

Table 11 looks at the token hit-rate for the models **BaselineBig** and **CodeswitchMonoGatiPanlexBig**, for the categories of animals occurring in GATITOS (*bear, bee, bird, butterfly, cat, chicken, deer, dog, elephant, fish, frog, goat, horse, insect, lion, monkey, parrot, pig, rabbit, sheep, snail, snake, tiger, turkey, turtle*), animals NOT appearing in GATITOS (*ant, antelope, buffalo, cheetah, crocodile, dolphin, dormouse, gorilla, jellyfish, koala, leopard, moose, mosquito, newt, ocelot, otter, reindeer, robin, scorpion, shark, sloth, spider, springbok, tortoise, velociraptor*), colors (*black, white, red, blue, yellow, green, purple, orange, grey*), and numbers (*one, two, three, four, five, six, seven, eight, nine, ten, hundred, million*). All numbers and colors appear in GATITOS. Numbers are included as a weak control, since the model will

tend to make fewer mistakes on them, though such UNMT-style mistakes do occur.

As expected, the GATITOS-augmented model performs better on these tokens. Two things are worth noting. First, the model improves noticeably on the complementary distribution—words that do not appear in the lexicon training data—but unsurprisingly improves more on the words that are present in GATITOS. Second, the improvements are not as large as expected: why is it not now getting 100% accuracy? Digging into the outputs, it seems that this is mainly due to the high baseline of (a) undertranslation; (b) hallucination; and (c) copying, as we expect from a model trained without various other tricks like back-translation (see Section G for an analysis of common error types). This point is underscored by the models’ imperfect performance on the “easy” class of numbers.

F Biggest winners

We also look at the top 5 languages that were the biggest gainers over the baseline for each model. In some cases these may represent remarkable successes of a particular approach—though in other cases they may represent noisy outliers, as is to be expected when evaluating 200 languages.

F.1 FLORES-200

en→xx The biggest winners for each model in the en→xx direction for the FLORES-200 evaluation set are given in Table 12.

There are seven languages that gained at least 5 CHRF over the baseline on at least one model trained with data augmentation. These languages are:

1. Bhojpuri (bho): up to +14.5 CHRF
2. Ilocano (ilo): up to +9.1 CHRF
3. Serbian (sr): up to +8.3 CHRF
4. Bambara (bm): up to +8.1 CHRF
5. Tibetan (bo): up to +8.0 CHRF

cat.	FLORES-200				GATONES			
	$A \in \text{GAT}$	$A \notin \text{GAT}$	colors	#s	$A \in \text{GAT}$	$A \notin \text{GAT}$	colors	#s
BaselineBig	36.6	36.6	55.3	63.3	33.4	25.8	35.4	53.5
CodeswitchMonoGatiPanlexBig	45.5	40.7	66.5	66.8	44.9	32.5	47.8	58.0
Δ	+8.9	+4.0	+11.1	+3.6	+11.5	+6.7	+12.4	4.5

Table 11: Comparing token hit-rate on classes of nouns known to have issues for UNMT models, along with a weak control of numbers. There are large improvements for animals in the GATITOS training lexicon ($A \in \text{GAT}$), as well as their complementary distribution, animals not in GATITOS ($A \notin \text{GAT}$), and colors. Number hit-rate has a minor bump.

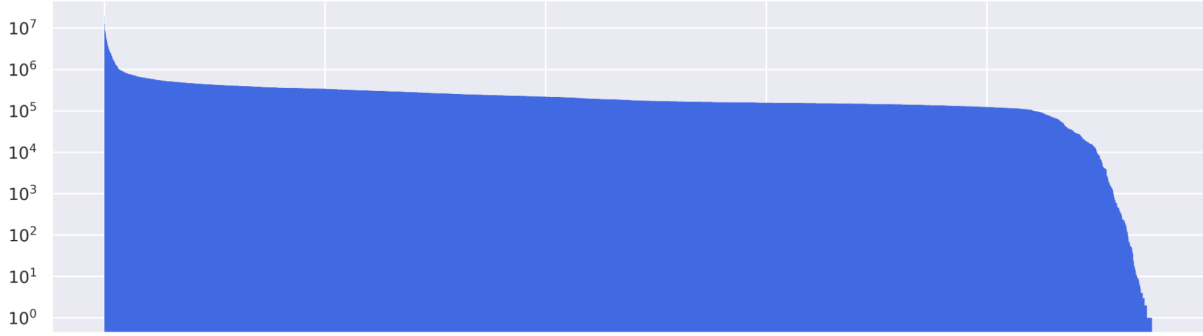


Figure 3: Number of word pairs available in Panlex for each of 4750 BCP-47 languages (log scale).

6. Nuer (nus): up to +6.8 CHRF

7. Mizo (lus): up to +6.2 CHRF

Unsurprisingly, most of these languages are unsupervised or low-resource, except for Serbian which is medium-resource in our dataset. Of the seven languages listed above, we use Panlex data for Ilocano, Serbian, Bambara, Tibetan, Nuer, and Mizo, and there is GATITOS data for Bhojpuri, Ilocano, Bambara, and Mizo. As will be discussed in Section 8.4, the GATITOS bilingual lexica are clearly a very useful resource for MT, although evidently Panlex alone can help as well. Another interesting finding is that Nuer, which has *no* English-aligned entries in Panlex but $\approx 20\text{K}$ non-English-aligned entries, still sees large improvements when translating from English. This is evidence that lexicon data can improve performance even in the zero-shot case, where e.g. the model learns better vocabulary alignment between English and Nuer despite not receiving explicit alignment information during training. In Section 8.4, we look at the relationship between the number of lexical data points for a language and the CHRF improvement, which provides some insight (albeit not perfect clarity) into why these particular languages did well.

xx $\rightarrow$ en Table 13 shows the top 5 biggest winners per model for the xx $\rightarrow$ en direction. Clearly there is a lot of overlap with the en $\rightarrow$ xx direction, although there are some differences. Also, note

again that the magnitude of improvement in this direction is smaller, likely because the baseline performance is higher and there is less improvement to be made simply by better aligning vocabulary cross-linguistically. Some of the biggest winners in this direction that weren’t already discussed for the en $\rightarrow$ xx direction are:

1. Tsonga (ts): up to +3.1 CHRF
2. Guarani (gn): up to +2.8 CHRF
3. Bashkir (ba): up to +2.5 CHRF
4. Minangkabau (min): up to +2.5 CHRF

F.2 GATONES

en $\rightarrow$ xx The biggest winners on GATONES in the en $\rightarrow$ xx direction are given in Table 14. Though there is some overlap with the biggest winners on the FLORES-200 dataset (e.g. Ilocano, Bambara, Mizo, Bhojpuri), a number of different languages perform well too, some of which simply aren’t included in the FLORES-200 set. The languages which gain > 5.0 CHRF on this part of the evaluation are:

1. Adyghe (ady): up to +14.1 CHRF
2. Kedah Malay (meo): up to +12.6 CHRF
3. Goan Konkani (gom): up to +11.6 CHRF
4. Bhojpuri (bho): up to +10.4 CHRF
5. Ilocano (ilo): up to +9.8 CHRF
6. Avar (av): up to +9.5 CHRF
7. Bambara (bm): up to +9.0 CHRF
8. Mizo (lus): up to +8.9 CHRF

Model					
GatiPanlexTokenPairs	ilo (+7.5)	nus (+6.8)	lus (+6.2)	bm (+5.5)	ts (+4.4)
CodeswitchMono	bho (+8.6)	bo (+5.6)	lus (+5.2)	ilo (+5.0)	quy (+4.9)
CodeswitchParallel	ilo (+3.7)	bho (+3.5)	as (+2.4)	lus (+2.3)	min (+1.9)
CodeswitchMonoParallel	bho (+9.4)	ilo (+6.4)	bo (+6.2)	lus (+6.0)	ln (+4.9)
CodeswitchMonoGatiPanlex	ilo (+9.1)	sr (+8.3)	bm (+8.1)	bho (+7.9)	bo (+6.8)
GlowupMono	sr (+6.2)	acq (+4.5)	bo (+3.8)	aeb (+3.1)	bho (+3.1)
GlowupParallel	bho (+4.4)	shn (+3.8)	sg (+3.7)	kac (+3.6)	kpb (+3.5)
GlowupMonoParallel	bho (+12.3)	sr (+7.1)	sg (+4.5)	bo (+4.1)	nus (+3.3)
GlowupMonoGatiPanlex	bho (+14.5)	bo (+8.0)	sr (+7.3)	bm (+5.7)	acq (4.2)

Table 12: Top 5 biggest winners per model (en→xx) on the FLORES-200 test set, measured in Δ CHRF over the baseline.

Model					
GatiPanlexTokenPairs	bm (+3.5)	ts (+3.1)	lus (+2.7)	gn (+2.6)	tum (+2.1)
CodeswitchMono	bo (+2.5)	lus (+2.4)	ti (+1.4)	ko (+1.3)	ay (+1.3)
CodeswitchParallel	mni (+2.2)	min (+1.9)	kg (+1.7)	lus (+1.4)	kbp (+1.4)
CodeswitchMonoParallel	bo (+2.2)	lus (+1.9)	ay (+1.4)	bm (+1.1)	ff (+1.0)
CodeswitchMonoGatiPanlex	lus (+5.0)	bo (+3.5)	bm (+3.2)	gn (+2.8)	ba (+2.5)
GlowupMono	bo (+2.4)	ti (+2.2)	ks (+2.2)	am (+2.0)	mai (+2.0)
GlowupParallel	min (+2.1)	ee (+1.9)	kg (+1.6)	kac (+1.5)	kbp (+1.5)
GlowupMonoParallel	lus (+2.9)	min (+2.5)	bo (+1.8)	ace (+1.8)	bug (+1.8)
GlowupMonoGatiPanlex	lus (+2.2)	gn (+2.2)	ko (+1.9)	sa (+1.8)	ckb (+1.7)

Table 13: Top 5 biggest winners per model (xx→en) on the FLORES-200 test set, measured in Δ CHRF over the baseline.

9. Madurese (mad): up to +6.6 CHRF
10. Assamese (as): up to +6.3 CHRF
11. Pattani Malay (mfa): up to +5.7 CHRF
12. Kalaallisut (kl): up to +5.4 CHRF
13. Tibetan (bo): up to +5.2 CHRF

xx→en The biggest winners in the xx→en direction are given in Table 15. Many of the biggest winners overlap with the en→xx direction, but some of the languages that haven’t yet been mentioned are:

1. Manipuri (mni{-Mtei}): up to +7.7 CHRF
2. Dogri (doi): up to +5.4 CHRF
3. Dhivehi (dv): up to +3.4 CHRF
4. Tigrinya (ti): up to +3.0 CHRF

F.3 Big model winners

The biggest winners for the 1.6B parameter models are given in Tables 16, 17, 18, and 19.

F.4 Empirical study of the quality of Panlex

One way to judge the quality of a dataset is to review it manually, as in (Kreutzer et al., 2022); another is to see the empirical effects on model quality of training on it. As a byproduct of using Panlex for this project, can we judge the quality of Panlex for different languages?

To reduce noise, we average the scores on the three main uses of the billexes, namely the TokenPairs model, the Glowup model, and the Codeswitch model. We average the FLORES-200 and GATONES scores. We then compare those scores to the baseline model for both en→xx and xx→en. For the purposes of this analysis, we treat any absolute delta of under 0.3 CHRF to be noise. The result is displayed in Tables 20 and 21.

One would like to say that the upper left-hand corner represents languages with unequivocally high-quality lexical data, and the lower right-hand corner represents languages with poor quality lexical data. Alas, however, this picture is rather muddled when we scale up to larger models, as we see that many languages jump from one bucket to another. Nonetheless, we do see the trend that GATITOS languages tend to cluster to the upper left-hand corner in both cases, and that Shan (‘shn’) and Latin (‘la’) do poorly in all cases, and should likely be avoided by practitioners.

Teasing out the confound of the mixed GATITOS and Panlex data: For the 26 GATITOS languages, it is harder to trust the previous analysis. However, we can compare the scores of these lan-

guages between the GatiPanlexTokenPairs model and the GatiTokenPairs model. The second of these models is trained on a strict subset of the data that the first is. If a language performs better with this subset of the data, we can presume that the Panlex data was on average lower quality; if a language performs better on the superset, the Panlex data might still be lower quality, but its quantity at least makes up for performance to some degree. The languages that do over +0.3 CHRF better on the subset data are ts, dv, bm, lus, ff, and ckb, suggesting that those may have poorer-quality Panlex data, with the largest difference being lus at +2.7 CHRF; those that do better on the superset are gom, mni-Mtei, kri, ln, doi, ay, sa, ti, mai, and as, suggesting that Panlex still adds useful signal there.

The picture that begins to come together is that Panlex often has some useful signal, but also contains considerable amounts of noise. For a less expressive model that is already not able to reach very high quality, some noise in the lexicons does not hurt, and Panlex can help the model get off the ground for the lowest-resource languages. But for a stronger baseline model that produces higher-quality translations on average, this noise can actively harm performance. Therefore, more carefully curated bilingual lexica, like GATITOS, will tend to will yield higher quality results when used for model training with bigger models, as evinced in Table 21.

Model					
GatiPanlexTokenPairs	gom (+11.6)	ilo (+8.1)	meo (+7.7)	bm (+6.8)	lus (+6.6)
CodeswitchMono	ady (+14.1)	av (+9.5)	meo (+6.8)	mad (+6.3)	gom (+5.3)
CodeswitchParallel	tiv (+3.6)	min (+3.4)	as (+3.3)	iso (+3.2)	mfa (+2.9)
CodeswitchMonoParallel	lus (+7.7)	ady (+6.8)	mad (+6.6)	bm (+6.0)	mfa (+5.7)
CodeswitchMonoGatiPanlex	bho (+10.4)	gom (+9.9)	ilo (+9.8)	bm (+9.0)	lus (+8.9)
GlowupMono	meo (+10.4)	bho (+6.6)	bo (+5.2)	gom (+5.0)	mfa (+4.1)
GlowupParallel	bal (+4.1)	yua (+3.1)	meo (+3.0)	tiv (+2.9)	mni (+2.6)
GlowupMonoParallel	meo (+7.1)	gom (+5.9)	mad (+3.9)	za (+3.7)	mni (+3.5)
GlowupMonoGatiPanlex	meo (+12.6)	gom (+11.6)	as (+6.3)	ilo (+6.2)	kl (+5.4)

Table 14: Top 5 biggest winners per model (en→xx) on the GATONES test set, measured in Δ CHRF over the baseline.

Model					
GatiPanlexTokenPairs	bm (+4.7)	mni-Mtei (+4.7)	gn (+3.8)	lus (+3.6)	ilo (+3.5)
CodeswitchMono	av (+3.5)	mni-Mtei (+2.9)	yua (+2.5)	dv (+2.3)	lus (+2.2)
CodeswitchParallel	mni (+3.0)	cv (+2.5)	av (+2.1)	lus (+1.9)	ee (+1.8)
CodeswitchMonoParallel	mni-Mtei (+7.7)	av (+3.8)	lus (+2.4)	bm (+1.5)	chr (+1.5)
CodeswitchMonoGatiPanlex	mni-Mtei (+5.4)	lus (+5.3)	gn (+3.5)	bm (+3.5)	dv (+3.4)
GlowupMono	ti (+3.0)	bo (+1.8)	or (+1.6)	quc (+1.4)	dv (+1.4)
GlowupParallel	ee (+2.2)	mad (+1.6)	yua (+1.6)	av (+1.6)	bm (+1.5)
GlowupMonoParallel	mad (+2.8)	min (+2.2)	lus (+2.0)	quc (+1.6)	gom (+1.3)
GlowupMonoGatiPanlex	mni-Mtei (+6.0)	doi (+5.4)	gom (+3.5)	dv (+3.3)	bm (+2.7)

Table 15: Top 5 biggest winners per model (xx→en) on the GATONES test set, measured in Δ CHRF over the baseline.

Model					
GatiPanlexTokenPairsBig	ts (+7.5)	din (+6.0)	ln (+ 5.8)	ilo (+5.3)	ay (+4.1)
CodeswitchMonoGatiPanlexBig	ts (+6.9)	bm (+5.6)	ilo (+5.2)	ln (+4.8)	mag (+3.8)

Table 16: Top 5 biggest winners per model (en→xx) on the FLORES-200 test set for the 1.6B parameter models, measured in Δ CHRF over the baseline.

Model					
GatiPanlexTokenPairsBig	tpi (+9.1)	mni (+6.2)	bm (+5.5)	ts (+3.5)	ay (+3.1)
CodeswitchMonoGatiPanlexBig	tpi (+5.1)	ay (+3.0)	mni (+2.2)	bm (+1.9)	ltg (+1.2)

Table 17: Top 5 biggest winners per model (xx→en) on the FLORES-200 test set for the 1.6B parameter models, measured in Δ CHRF over the baseline.

Model					
GatiPanlexTokenPairsBig	ts (+7.7)	gom (+6.1)	ilo (+6.0)	dv (+5.8)	bm (+5.1)
CodeswitchMonoGatiPanlexBig	ts (+7.3)	bm (+7.0)	ilo (+6.8)	mni-Mtei (+5.5)	gom (+5.3)

Table 18: Top 5 biggest winners per model (en→xx) on the GATONES test set for the 1.6B parameter models, measured in Δ CHRF over the baseline.

Model					
GatiPanlexTokenPairsBig	cv (+8.7)	mni (+8.4)	bm (+6.6)	kl (+5.8)	ee (+5.0)
CodeswitchMonoGatiPanlexBig	cv (+6.9)	kl (+6.2)	ce (+3.4)	av (+2.8)	chr (+2.7)

Table 19: Top 5 biggest winners per model (xx→en) on the GATONES test set for the 1.6B parameter models, measured in ΔCHRF over the baseline.

	Win xx→en	Neut. xx→en	Loss xx→en
Win en→xx	aa ace av ay bci bm bo cv doi dv dyu dz ee gn gom ilo kbp kl kri lg lus mad min mni - Mtei nso nus om quc quy sg ts yua	cjk ckb ny ti tn	acm acq aeb af am apc ar ar-MA arz as awa az ba bbc be bg bho bn bs ca ce ceb cs cy da de el eo et eu fa-AF fi fil fj fo fr ga gl gu hr hu hy id is iso it iw ja jv ka kk km kn ko ks ku lb lo lt ltg lv mag mfa mg mk ml mn mni mr ms mt nl no pa pt rn ro ru rw scn si sk sl sn so sq sr su sv sw ta te tg th tr tt uk ur uz vi war xh yue zh zu
Neut. en→xx	ak bug pag qu tpi	bew kmb or pcm sat-Beng sm	ban brx-Beng es fa gd ha hi ht ig ky ln mai mi my ne oc pap pl ps sd skr st tk ug vec wo yi yo zza
Loss en→xx	din ff fon kg sa tum	ady ber kac	ahr ber-Latn hne la shn

Table 20: Languages sorted by whether it helps or hurts to include PanLex and GATITOS, for the smaller models (Transformer Big, 475M). GATITOS languages bolded.

	Win xx→en	Neut. xx→en	Loss xx→en
Win en→xx	ay bm cv dv ee gom ilo kl kri ltg mni - Mtei sa	awa bci din doi fo ga lus mag quy tt	aeb ahr ak ba ckb dyu kg lg ln mfa pag pap rn sat-Beng skr su ts yua
Neut. en→xx	ce km mg tpi zza	af am ar az ban be bg bho bn bs ca cs cy da de el eo es et eu fa-AF ff fi fil fr gd gl gu ha hi hne hr ht hu hy id is it iw ja ka kk kn ko ky lb lo lt lv mad mi mk ml mn mr ms mt ne nl no ny pa pcm pl pt qu ro ru si sk sl so sq sr sv sw ta te tg th tk tr uk ur uz vi yi yo zh zu	ace acm acq apc ar-MA arz bew brx-Beng bug ceb dz fj ig jv ks nso oc om or ps scn sd sm sn st ug vec xh yue
Loss en→xx	ady av ber-Latn min mni	aa bbc cjk gn kmb my sg tn tum	as ber bo fon iso kac kbp ku la mai nus quc rw shn ti war wo

Table 21: Languages sorted by whether it helps or hurts to include PanLex and GATITOS, for the bigger models (Transformer 1.6B). GATITOS languages bolded.

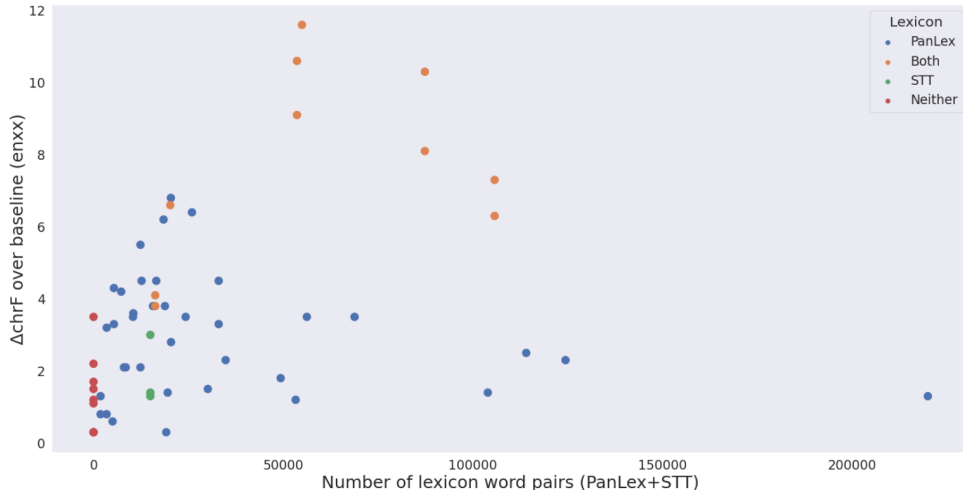


Figure 4: Number of lexicon word pairs in augmented data vs. ΔChrF over baseline for unsupervised languages in the $\text{en} \rightarrow \text{xx}$ direction. Results for FLORES-200 and GATONES are combined here.

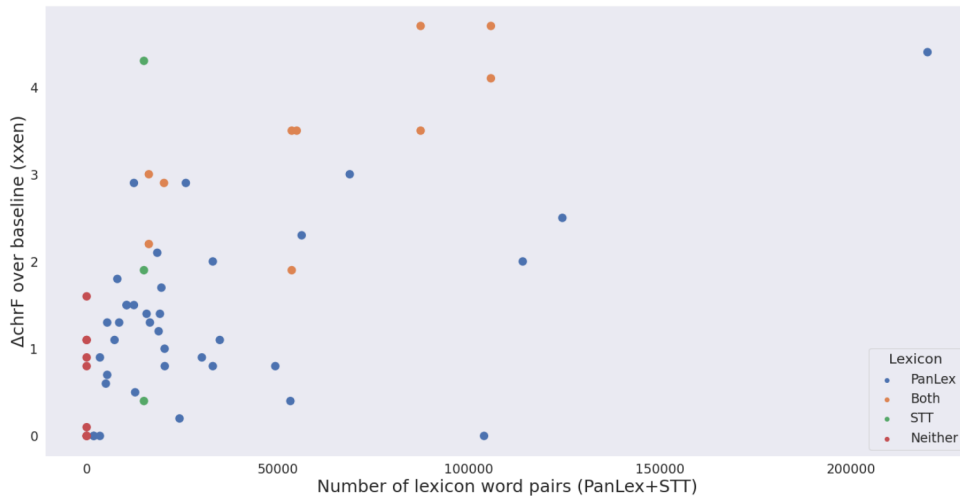


Figure 5: Number of lexicon word pairs in augmented data vs. ΔChrF over baseline for unsupervised languages in the $\text{xx} \rightarrow \text{en}$ direction. Results for FLORES-200 and GATONES are combined here.

G Does lexical augmentation fix common MT mistakes?

In evaluating the “big” models with 1.6B parameters, we wished to see whether our preferred lexical data augmentation methods (GatiPanlexTokenPairs or CodeswitchMonoGatiPanlex) reduced several common types of MT errors. The errors we looked at were (1) null output, or the “question mark phenomenon,” where the model simply outputs some unrelated symbol (such as question marks) instead of actual text; (2) copying, where the model copies some or all of the source sentence in its prediction; and (3) repetition, where the model erroneously repeats the same word or phrase many times. There are other error types we could look at, like hallucination, but we stick with these three basic types for this paper. More precise definitions of these errors

are given below.

The results of this analysis are given in Table 22. For each error type, we computed the percentage of sentences that were affected by dividing the number of affected sentences by the total number of sentences in the eval set. FLORES-200 has 806248 sentence pairs across all languages and GATONES has 309887.

Null output (question mark phenomenon) The first error type occurs when the model outputs only “??”, or some other arbitrary character, as its prediction. Instances of this likely indicate catastrophic effects of out-of-domain phenomena for 0-shot translation.

Copying Another common error is copying, where the model’s prediction is close or identi-

	% question marks	% near copy	% repetition
FLORES-200			
BaselineBig	0.7	4.5	3.4
GatiPanlexTokenPairsBig	0.9	3.6	2.7
CodeswitchMonoGatiPanlexBig	0.4	4.7	3.2
GATONES			
BaselineBig	0.6	2.2	3.1
GatiPanlexTokenPairsBig	0.6	1.8	2.2
CodeswitchMonoGatiPanlexBig	0.4	2.3	2.6

Table 22: The frequencies of three common error types in MT in each of the eval sets, as a percentage of the total sentences in the set that had each issue (lower is better). Exact descriptions of the error types are given in Section G.

cal to the source sentence. In measuring this phenomenon, we said that any prediction with $> 85\%$ character-level similarity to the source sentence was considered a copy. To measure character-level similarity, we took the multiset intersection of the character frequencies in the source and the character frequencies in the prediction, and then divided the size of the intersection by the number of characters in the source.

Repetition The last common error type we examined was repetition. To count these mistakes, we divided the total number of tokens in a sentence by the number of *unique* tokens. If the ratio was > 3 , we counted the prediction as an instance of erroneous repetition.

H Comparing sampling strategies for translating tokens

As one recalls from Section 6.1.1, the Codeswitch augmentation works as follows: Let D represent a multilingual lexicon containing word or phrase translation pairs for many languages. Given a source sentence $x = (x_1, x_2, \dots, x_n)$ from monolingual corpus X_{mono} , we substitute each token in x for its dictionary translation with probability p_{tr} .

However, there is an issue with this formulation. Because the lexica we use do not have exhaustive coverage across languages, it is often the case that simply looping over x and attempting to translate each token with probability p_{tr} would result in translating a fraction of x that is significantly less than p_{tr} . So in order to approximate this desired fraction p_{tr} as closely as possible, we first count how many tokens in x have dictionary translations. Let this number be k . We then compute the adjusted probability $\tilde{p}_{tr} = \max(\frac{np_{tr}}{k}, 1)$, and sample from amongst the words in x with translations

with probability $\tilde{p}_{tr}$, to obtain the codeswitched sentence x' . When substituting a source word for its translation, we choose a translation uniformly at random from all available translations in all languages. Because of this, it is usually the case that x is codeswitched into many languages. Finally, we train the model to reconstruct the monolingual sentence x from x' using the same sequence-to-sequence model and loss function as for the MT task.

In our experiments we use $p_{tr} = 0.4$. We apply MCA to all 208 languages in our corpus, but augment only half the available monolingual data and train the remaining half with MASS (Song et al., 2019b), as done in the baseline training regime (Bapna et al., 2022). We prepend a task token, `<2codeswitch>`, to the codeswitched sentences to cue the model to perform the MCA task, as well as language (`<2lang>`) and script (`<2script>`) tokens. The `<2lang>` and script `<2script>` tags are used in all models, including the baseline.

Since this augmentation samples each token with some probability, the number of tokens translated in a given sentence follows a binomial distribution. The Glowup augmentation, however, samples a number of tokens to translate uniformly at random from all possible translatable tokens. So one has a binomial distribution over N tokens sampled, and the other has a uniform distribution—does this make a difference?

To test this we trained a version of the CodeswitchMono model using uniform sampling. The average CHRF of the CodeswitchMonoUniform model was 0.1 to 0.2 higher on all four of the (en $\rightarrow$ xx xx $\rightarrow$ en) x (FLORES-200 GATONES) directions. We conclude that this may have a slight benefit, but the difference is within the realm of

noise, and does not affect the conclusions elsewhere in this paper.

I Relationship between number of tokens and MT performance

We also graph the relationship between number of lexical word pairs and ΔCHRF in Figures 4 (en $\rightarrow$ xx) and 5 (xx $\rightarrow$ en) for URLs only. The results for FLORES-200 and GATONES are combined in these plots. In both directions, we observe a moderate positive relationship between the number of lexical word pairs for a given language in the augmented data and the ΔCHRF over the baseline.

J Languages

J.1 Rationale for Language Choice

Although this project is aligned with the 1000-language initiative from Bapna et al. (2022), we wanted to use smaller models for more rapid iteration, and as a result, commensurately smaller data and number of languages to fit comfortably in the model. Therefore, we chose to work with about 200 languages.

With this in mind, we also wanted to choose specifically those languages whose performance we could measure. Therefore, our approach was as follows:

- Include all languages with supervised (parallel) data, for maximal cross-lingual transfer
- Include all languages that have non-zero data and a FLORES-200 eval set
- Include all languages that have non-zero data and a GATONES eval set

J.2 Complete Language data

The following table gives a list of the languages used in our experiments, along with some linguistic and resource-related statistics. The numbers for data resources (i.e. Mono, Parallel, Panlex, and GATITOS) refer to the amount of data actually used in our experiments, *not* necessarily the total amount of data available. For example, we subsampled the parallel and high-resource monolingual data we had available by a factor of 10.

BCP-47	Language	Cat.	Mono	Parallel	PanLex	GATITOS	Speak.	Script	Cont.	Family
en	English	HRL	738.8M	4726.4M	3.6M	0	984M	Latn	Europe	Indo-European
es	Spanish	HRL	175.1M	585.6M	2.4M	4K	528M	Latn	Europe	Indo-European
de	German	HRL	169.3M	389.3M	2.8M	4K	130M	Latn	Europe	Indo-European
id	Indonesian	HRL	95M	97.4M	1.1M	4K	198M	Latn	Asia	Austronesian
yue	Cantonese	MRL	83.1M	405K	180K	0	84M	Hant	Asia	Sino-Tibetan
hu	Hungarian	HRL	72.4M	50.9M	1.3M	0	13M	Latn	Europe	Indo-European
pl	Polish	HRL	68.7M	152.8M	1.4M	4K	41M	Latn	Europe	Indo-European
zh	Mandarin	HRL	67.6M	215.7M	6K	4K	1092M	Hans	Asia	Sino-Tibetan
mi	Maori	MRL	67.4M	1.3M	474K	0	50K	Latn	Oceania	Austronesian
ko	Korean	HRL	67.2M	128.1M	1.3M	5K	77M	Kore	Asia	Koreanic
ja	Japanese	HRL	65.2M	307.7M	2.2M	4K	128M	Jpan	Asia	Japonic
lo	Lao	MRL	59.4M	817K	184K	0	30M	Laoo	Asia	Kra-Dai
ru	Russian	HRL	57M	294.7M	2.8M	4K	268M	Cyrl	Europe	Indo-European
gd	Scottish Gaelic	MRL	56.9M	4M	324K	0	57K	Latn	Europe	Indo-European
tr	Turkish	HRL	56.5M	159.2M	1.3M	4K	71M	Latn	Asia	Turkic
so	Somali	MRL	56.4M	1.3M	112K	0	16M	Latn	Africa	Afro-Asiatic
th	Thai	HRL	53.1M	69.1M	1.6M	4K	61M	Thai	Asia	Kra-Dai
ha	Hausa	MRL	50.6M	1.8M	202K	0	80M	Latn	Africa	Afro-Asiatic
it	Italian	HRL	49.1M	245.5M	2M	4K	66M	Latn	Europe	Indo-European
pt	Portuguese	HRL	48.5M	240.7M	1.7M	4K	230M	Latn	Europe	Indo-European
vi	Vietnamese	HRL	47.7M	94.2M	825K	4K	68M	Latn	Asia	Austroasiatic
fr	French	HRL	45M	481.6M	2.5M	4K	230M	Latn	Europe	Indo-European
ceb	Cebuano	MRL	43.9M	9.2M	62K	0	20M	Latn	Asia	Austronesian
yo	Yoruba	MRL	43.4M	847K	244K	0	50M	Latn	Africa	Niger-Congo
sd	Sindhi	MRL	42.3M	1.6M	43K	0	26M	Arab	Asia	Indo-European
co	Corsican	MRL	41.2M	1.5M	148K	0	150K	Latn	Europe	Indo-European
mg	Malagasy	MRL	40.2M	3.6M	116K	0	25M	Latn	Africa	Austronesian
ms	Malay	HRL	39.4M	53.8M	0	0	77M	Latn	Asia	Austronesian
ar-MA	Mor. Arabic	URL	35.5M	0	0	0	52M	Arab	Africa	Afro-Asiatic
bew	Betawi	URL	33.3M	0	5K	0	5M	Latn	Asia	Malay Creole
ny	Nyanja	MRL	32.9M	1.2M	26K	0	12M	Latn	Africa	Niger-Congo
nl	Dutch	HRL	32M	258M	1.6M	4K	22M	Latn	Europe	Indo-European
uk	Ukrainian	HRL	32M	75.3M	892K	0	35M	Cyrl	Europe	Indo-European
sv	Swedish	HRL	31.1M	122.6M	1.3M	0	12M	Latn	Europe	Indo-European
haw	Hawaiian	MRL	30.4M	698K	156K	0	24K	Latn	Americas	Austronesian
ro	Romanian	HRL	30.1M	45M	841K	0	24M	Latn	Europe	Indo-European
cs	Czech	HRL	30M	106.9M	1.6M	0	13M	Latn	Europe	Indo-European
hmn	Hmong	MRL	27.7M	4.9M	4K	0	4M	Latn	Europe	Hmong-Mien
yi	Yiddish	MRL	27.6M	760K	0	0	2M	Hebr	Europe	Indo-European
fa	Persian	HRL	27.5M	45.2M	0	0	53M	Arab	Asia	Indo-European
ig	Igbo	MRL	27.4M	647K	48K	0	27M	Latn	Africa	Niger-Congo
lv	Latvian	MRL	26M	22.4M	0	0	2M	Latn	Europe	Indo-European
ar	Arabic	HRL	25.8M	116.7M	0	4K	310M	Arab	Asia	Afro-Asiatic
ckb	Sorani	MRL	25.1M	155K	53K	4K	7M	Arab	Asia	Indo-European
tt	Tatar	MRL	25M	557K	128K	0	5M	Cyrl	Europe	Turkic
sm	Samoan	MRL	23.9M	502K	58K	0	510K	Latn	Oceania	Austronesian
zu	Zulu	MRL	23.4M	2.3M	144K	0	12M	Latn	Africa	Niger-Congo
no	Norwegian	HRL	23.1M	85.8M	4K	0	5M	Latn	Europe	Indo-European
st	Sesotho	MRL	22.6M	1.2M	35K	0	6M	Latn	Africa	Niger-Congo
ta	Tamil	MRL	22M	11.1M	247K	0	76M	Taml	Asia	Dravidian
or	Odia (Oriya)	MRL	21.2M	169K	16K	0	35M	Orya	Asia	Indo-European
sn	Shona	MRL	18.5M	958K	102K	0	8M	Latn	Africa	Niger-Congo
bo	Tibetan	LRL	17.8M	282K	56K	0	1M	Tibt	Asia	Sino-Tibetan
el	Greek	HRL	17.3M	54M	1.2M	0	13M	Grek	Europe	Indo-European
fi	Finnish	HRL	17.1M	48.6M	2.1M	0	6M	Latn	Europe	Uralic
hi	Hindi	HRL	16.5M	75.7M	449K	4K	381M	Deva	Asia	Indo-European
xh	Xhosa	LRL	15.8M	697K	57K	0	8M	Latn	Africa	Niger-Congo
mr	Marathi	MRL	15.6M	8.1M	154K	0	75M	Deva	Asia	Indo-European
sk	Slovak	HRL	15.6M	63.9M	1.1M	0	7M	Latn	Europe	Indo-European
hy	Armenian	MRL	15.4M	6.9M	751K	0	5M	Armn	Asia	Indo-European
kk	Kazakh	MRL	15.3M	6.6M	241K	0	13M	Cyrl	Asia	Turkic
da	Danish	HRL	15.2M	78.9M	545K	0	6M	Latn	Europe	Indo-European
mk	Macedonian	MRL	15M	6.6M	363K	0	2M	Cyrl	Europe	Indo-European
bg	Bulgarian	MRL	14.9M	37.4M	693K	0	8M	Cyrl	Europe	Indo-European
sr	Serbian	MRL	14.1M	30.6M	206K	0	8M	Cyrl	Europe	Indo-European
ml	Malayalam	MRL	13.9M	6.4M	163K	0	34M	Mlym	Asia	Dravidian
az	Azerbaijani	MRL	13.5M	19.6M	0	0	23M	Latn	Asia	Turkic
is	Icelandic	MRL	13.4M	15.8M	620K	0	310K	Latn	Europe	Indo-European
te	Telugu	MRL	12.7M	8M	311K	0	79M	Telu	Asia	Dravidian
ne	Nepali	MRL	11.6M	9.7M	0	0	16M	Deva	Asia	Indo-European
mzn	Mazanderani	URL	11.6M	0	29K	0	6M	Arab	Asia	Indo-European
meo	Kedah Malay	URL	11.3M	0	0	0	3M	Latn	Asia	Austronesian
et	Estonian	MRL	11.2M	30.1M	0	0	1M	Latn	Europe	Uralic
iw	Hebrew	HRL	10.8M	57.1M	707K	0	5M	Hebr	Asia	Afro-Asiatic
rw	Kinyarwanda	LRL	10.1M	803K	56K	0	10M	Latn	Africa	Niger-Congo
mn	Mongolian	LRL	10M	4.1M	0	0	5M	Cyrl	Asia	Mongolic
ur	Urdu	MRL	9.9M	15.2M	400K	0	163M	Arab	Asia	Indo-European
apc	N. Lev. Arabic	URL	9.7M	0	0	0	15M	Arab	Asia	Afro-Asiatic
hr	Croatian	MRL	9.7M	17.4M	738K	0	7M	Latn	Europe	Indo-European
fil	Filipino	MRL	9.5M	25.3M	61K	0	45M	Latn	Asia	Austronesian
as	Assamese	LRL	9.3M	575K	78K	4K	15M	Beng	Asia	Indo-European
arz	Egyptian Arabic	URL	9.2M	0	101K	0	58M	Arab	Africa	Afro-Asiatic
fo	Faroese	LRL	9.2M	26K	254K	0	66K	Latn	Europe	Indo-European
pap	Papiamentu	URL	9.1M	0	99K	0	341K	Latn	Americas	Portuguese Creole
fa-AF	Dari	LRL	8.7M	10K	0	0	21M	Arab	Asia	Indo-European
acm	Mesop. Arabic	URL	8.7M	0	11K	0	15M	Arab	Asia	Afro-Asiatic

BCP-47	Language	Cat.	Mono	Parallel	PanLex	GATITOS	Speak.	Script	Cont.	Family
lt	Lithuanian	MRL	8.6M	30.9M	611K	0	3M	Latn	Europe	Indo-European
lus	Mizo	URL	8.3M	0	85K	4K	688K	Latn	Asia	Sino-Tibetan
be	Belarusian	LRL	8.3M	6.5M	508K	0	3M	Cyrl	Europe	Indo-European
kn	Kannada	LRL	8M	5.8M	116K	0	47M	Knda	Asia	Dravidian
dv	Dhivehi	LRL	7.9M	1K	34K	4K	300K	Thaa	Asia	Indo-European
my	Burmese	LRL	7.8M	5.5M	121K	0	43M	Mymr	Asia	Sino-Tibetan
oc	Occitan	LRL	7.5M	6K	2.4M	0	500K	Latn	Europe	Indo-European
bn	Bengali	MRL	7.3M	21.7M	259K	0	262M	Beng	Asia	Indo-European
af	Afrikaans	MRL	7.1M	12.7M	258K	0	18M	Latn	Africa	Indo-European
eu	Basque	LRL	7M	6.4M	792K	0	540K	Latn	Europe	Language isolate
gu	Gujarati	LRL	6.9M	5.8M	198K	0	47M	Gujr	Asia	Indo-European
gl	Galician	MRL	6.4M	13.1M	383K	0	2M	Latn	Europe	Indo-European
sa	Sanskrit	LRL	6.2M	11K	168K	4K	100K	Deva	Asia	Indo-European
sl	Slovenian	MRL	5.9M	27M	672K	0	2M	Latn	Europe	Indo-European
ug	Uyghur	LRL	5.7M	526K	116K	0	10M	Arab	Asia	Turkic
ba	Bashkir	LRL	5.6M	303K	112K	0	1M	Cyrl	Europe	Turkic
si	Sinhala	LRL	5.6M	6.5M	88K	0	16M	Sinh	Asia	Indo-European
om	Oromo	LRL	5.6M	203K	10K	4K	24M	Latn	Africa	Afro-Asiatic
zza	Zaza	URL	5.3M	0	0	0	2M	Latn	Asia	Indo-European
uz	Uzbek	MRL	5.3M	11.2M	0	0	34M	Latn	Asia	Turkic
sw	Swahili	MRL	5.2M	10.5M	0	0	150M	Latn	Africa	Niger-Congo
km	Khmer	MRL	5.1M	8M	188K	0	17M	Khmr	Asia	Austroasiatic
ky	Kyrgyz	LRL	4.9M	2.9M	174K	0	5M	Cyrl	Asia	Turkic
am	Amharic	LRL	4.7M	2.8M	71K	0	26M	Ethi	Africa	Afro-Asiatic
vec	Venetian	URL	4.4M	0	202K	0	4M	Latn	Europe	Indo-European
ca	Catalan	MRL	4.4M	32.9M	803K	0	9M	Latn	Europe	Indo-European
tk	Turkmen	LRL	4M	416K	177K	0	7M	Latn	Asia	Turkic
ti	Tigrinya	LRL	3.9M	67K	45K	4K	8M	Ethi	Africa	Afro-Asiatic
pa	Punjabi	LRL	3.7M	3.3M	89K	0	29M	Guru	Asia	Indo-European
sq	Albanian	MRL	3.2M	10.6M	269K	0	13M	Latn	Europe	Indo-European
ka	Georgian	MRL	3M	11.7M	482K	0	4M	Geor	Asia	Kartvelian
cv	Chuvash	URL	2.8M	0	121K	0	1M	Cyrl	Europe	Turkic
ilo	Ilocano	URL	2.6M	0	41K	4K	9M	Latn	Asia	Austronesian
bal	Baluchi	URL	2.5M	0	16K	0	8M	Arab	Asia	Indo-European
eo	Esperanto	LRL	2.4M	7.5M	1.3M	0	2M	Latn	Europe	Constructed
cy	Welsh	LRL	2.2M	6.4M	448K	0	590K	Latn	Europe	Indo-European
la	Latin	LRL	2.2M	2.2M	740K	0	0	Latn	Europe	Indo-European
dz	Dzongkha	LRL	2.1M	260K	31K	0	200K	Tibt	Asia	Sino-Tibetan
mt	Maltese	LRL	2.1M	7.3M	247K	0	470K	Latn	Europe	Afro-Asiatic
tn	Tswana	LRL	2M	66K	100K	0	8M	Latn	Africa	Niger-Congo
lg	Luganda	LRL	2M	3K	31K	4K	4M	Latn	Africa	Niger-Congo
ht	Haitian	LRL	2M	3.4M	177K	0	8M	Latn	Americas	French Creole
nso	Sepedi	LRL	1.9M	798K	13K	4K	5M	Latn	Africa	Niger-Congo
ps	Pashto	LRL	1.8M	2.1M	0	0	50M	Arab	Asia	Indo-European
za	Zhuang	URL	1.7M	0	0	0	15M	Latn	Asia	Kra-Dai
ga	Irish	LRL	1.7M	4M	428K	0	1M	Latn	Europe	Indo-European
tpi	Tok Pisin	LRL	1.7M	3.3M	62K	0	120K	Latn	Oceania	English Creole
pcm	Nigerian Pidgin	LRL	1.6M	24K	4K	0	40M	Latn	Africa	English Creole
lb	Luxembourgish	LRL	1.5M	4.6M	183K	0	420K	Latn	Europe	Indo-European
ku	Kurmanji	LRL	1.5M	2.1M	0	0	15M	Latn	Asia	Indo-European
tg	Tajik	LRL	1.4M	1.5M	194K	0	8M	Cyrl	Asia	Indo-European
ln	Lingala	LRL	1.4M	5K	97K	4K	58M	Latn	Africa	Niger-Congo
ce	Chechen	URL	1.4M	0	113K	0	1M	Cyrl	Europe	NE Caucasian
mai	Maithili	URL	1.3M	0	0	4K	65M	Deva	Asia	Indo-European
jav	Javanese	LRL	1.3M	6.2M	128K	0	84M	Latn	Asia	Austronesian
ts	Tsonga	LRL	1.3M	2K	9K	4K	13M	Latn	Africa	Niger-Congo
fj	Fijian	LRL	1.3M	6K	44K	0	339K	Latn	Oceania	Austronesian
ak	Twi	LRL	1.3M	38K	115K	4K	11M	Latn	Africa	Niger-Congo
ber-Latn	Tamazight	URL	1.2M	0	2K	0	30M	Latn	Africa	Afro-Asiatic
su	Sundanese	LRL	1.2M	2.7M	90K	0	34M	Latn	Asia	Austronesian
fy	Western Frisian	LRL	1.1M	4.8M	90K	0	850K	Latn	Europe	Indo-European
skr	Saraiki	URL	974K	0	0	0	20M	Arab	Asia	Indo-European
bbc	Batak Toba	URL	932K	0	23K	0	2M	Latn	Asia	Austronesian
war	Waray (PHs)	URL	902K	0	48K	0	3M	Latn	Asia	Austronesian
gn	Guarani	LRL	861K	1.3M	6K	4K	5M	Latn	Americas	Tupian
qu	Quechua	LRL	842K	2K	46K	4K	9M	Latn	Americas	Quechuan
bug	Buginese	URL	797K	0	19K	0	6M	Latn	Asia	Austronesian
ee	Ewe	LRL	796K	4K	90K	4K	4M	Latn	Africa	Niger-Congo
ltg	Latgalian	URL	796K	0	34K	0	170K	Latn	Europe	Indo-European
kl	Kalaallisut	LRL	741K	500	48K	4K	56K	Latn	Americas	Eskimo-Aleut
bho	Bhojpuri	LRL	734K	4K	0	4K	60M	Deva	Asia	Indo-European
ar-Latn	Arabic	URL	634K	0	4K	0	3M	Latn	Asia	Afro-Asiatic
pag	Pangasinan	URL	594K	0	19K	0	1M	Latn	Asia	Austronesian
shn	Shan	URL	566K	0	16K	0	3M	Mymr	Asia	Kra-Dai
min	Minangkabau	URL	533K	0	12K	0	6M	Latn	Asia	Austronesian
cjk	Chokwe	URL	494K	0	8K	0	983K	Latn	Africa	Niger-Congo
yua	Yucateco	URL	419K	0	67K	0	766K	Latn	Americas	Mayan
sg	Sango	URL	410K	0	32K	0	400K	Latn	Africa	Ngbandi Creole
iso	Isoko	URL	409K	0	3K	0	420K	Latn	Africa	Niger-Congo
kac	Kachin	URL	402K	0	10K	0	940K	Latn	Asia	Sino-Tibetan
kg	Kongo	LRL	376K	5K	15K	0	7M	Latn	Africa	Niger-Congo
gom	Goan Konkani	URL	311K	0	37K	4K	2M	Deva	Asia	Indo-European
bs	Bosnian	MRL	311K	22.6M	112K	0	2M	Cyrl	Europe	Indo-European
av	Avaric	URL	301K	0	216K	0	760K	Cyrl	Europe	Northeast Caucasian
tiv	Tiv	URL	297K	0	13K	0	2M	Latn	Africa	Niger-Congo
ady	Adyghe	URL	296K	0	25K	0	575K	Cyrl	Europe	Northwest Caucasian

BCP-47	Language	Cat.	Mono	Parallel	PanLex	GATITOS	Speak.	Script	Cont.	Family
wo	Wolof	LRL	289K	290K	94K	0	4M	Latn	Africa	Niger-Congo
hne	Chhattisgarhi	URL	269K	0	0	0	18M	Deva	Asia	Indo-European
ay	Aymara	LRL	267K	600	91K	4K	3M	Latn	Americas	Aymaran
quc	K'iche'	URL	250K	0	54K	0	2M	Latn	Americas	Mayan
ace	Achinese	URL	226K	0	18K	0	4M	Latn	Asia	Austronesian
acq	Mesop. Arabic	URL	216K	0	3K	0	7M	Arab	Asia	Afro-Asiatic
fon	Fon	URL	197K	0	8K	0	1M	Latn	Africa	Niger-Congo
ban	Balinese	LRL	188K	9K	30K	0	3M	Latn	Asia	Austronesian
bm	Bambara	URL	187K	0	73K	4K	14M	Latn	Africa	Mande
doi	Dogri	URL	179K	0	0	4K	2M	Deva	Asia	Indo-European
tum	Tumbuka	LRL	171K	4K	1K	0	2M	Latn	Africa	Niger-Congo
bci	Baoulé	URL	152K	0	20K	0	2M	Latn	Africa	Niger-Congo
quy	Ayacucho Quechua	URL	140K	0	94K	0	900K	Latn	Americas	Quechuan
mad	Madurese	URL	138K	0	18K	0	7M	Latn	Asia	Austronesian
awa	Awadhi	URL	136K	0	0	0	38M	Deva	Asia	Indo-European
dyu	Dyula	URL	130K	0	5K	4K	3M	Latn	Africa	Mande
kbp	Kabiyè	URL	129K	0	10K	0	1M	Latn	Africa	Niger-Congo
kri	Krio	URL	129K	0	10K	4K	496K	Latn	Africa	English Creole
rn	Rundi	LRL	125K	2K	27K	0	9M	Latn	Africa	Niger-Congo
mni	Manipuri	URL	106K	0	2K	0	1M	Beng	Asia	Sino-Tibetan
mni-Mtei	Manipuri	URL	103K	0	1K	4K	1M	Mtei	Asia	Sino-Tibetan
ber	Tamazight	URL	96K	0	0	0	30M	Tfng	Africa	Afro-Asiatic
kmb	Kimbundu	URL	94K	0	7K	0	4M	Latn	Africa	Niger-Congo
scn	Sicilian	URL	92K	0	149K	0	5M	Latn	Europe	Indo-European
ff	Fulah	LRL	86K	4K	30K	4K	50M	Latn	Africa	Niger-Congo
aa	Afar	URL	82K	0	52K	0	4M	Latn	Africa	Afro-Asiatic
ks	Kashmiri	LRL	71K	1K	24K	0	6M	Arab	Asia	Indo-European
mag	Magahi	URL	66K	0	0	0	14M	Deva	Asia	Indo-European
chr	Cherokee	LRL	63K	76K	61K	0	13K	Cher	Americas	Iroquoian
din	Dinka	URL	62K	0	3K	0	1M	Latn	Africa	Nilo-Saharan
aeb	Tunisian Arabic	URL	48K	0	13K	0	11M	Arab	Africa	Afro-Asiatic
ahr	Ahirani	URL	24K	0	0	0	2M	Deva	Asia	Indo-European
nus	Nuer	URL	24K	0	20K	0	890K	Latn	Africa	Nilo-Saharan
mfa	Pattani Malay	URL	7K	0	0	0	1000K	Arab	Asia	Austronesian
sat-Beng	Santali	URL	7K	0	0	0	6M	Beng	Asia	Austroasiatic
brx-Beng	Bodo (India)	URL	4K	0	0	0	1M	Beng	Asia	Sino-Tibetan

K Full results

The full results on FLORES-200 for the various models we trained are available in Tables 24 and 25. Model abbreviations are clarified in Table 23. Scores from the NLLB model are included as reference, though keep in mind that the smaller research models in this paper will naturally have lower quality; even the “Big” models are $30\times$ smaller, and not optimized with back-translation and so on.

Model Name	Abbr.	Description
Baseline	B	Trained on MASS + translation
GatiPanlexTokenPairs	T	Baseline + 5% token pairs
GatiPanlexTokenPairsSamp75	T_{75}	token-pairs sampled to 75%
GatiPanlexTokenPairsSamp50	T_{50}	token-pairs sampled to 50%
GatiPanlexTokenPairsSamp25	T_{25}	token-pairs sampled to 25%
GatiTokenPairs	T_{GAT}	token pairs with only GATITOS
CodeswitchMono	C_M	See §6
CodeswitchParallel	C_P	See §6
CodeswitchMonoParallel	C_{MP}	See §6
CodeswitchMonoGatiPanlex	$C_M T$	See §6
GlowupMono	G_M	See §6
GlowupParallel	G_P	See §6
GlowupMonoParallel	G_{MP}	See §6
GlowupMonoGatiPanlex	$G_M T$	See §6
BaselineBig	B_{BIG}	Big version of the Baseline (§8.1)
GatiPanlexTokenPairsBig	T_{BIG}	Big version of the TP (§8.1)
CodeswitchMonoGatiPanlexBig	$(C_M + T)_{BIG}$	Big version of $C_M T$ (§8.1)
NLLB	NL	NLLB 54B model

Table 23: Names, abbreviations, and descriptions of the full model results in Tables 25 and 24.

lang	cat.	B	T	T75	T50	T25	T _{Gatitos}	C _M	C _P	C _{MP}	C _M T	G _M	G _P	G _{MP}	G _M T	B _{BIG}	T _{BIG}	(C _M T) _{BIG}	NLLB
μ	mean	39.7	40.0	40.0	39.6	39.8	39.7	40.4	38.3	39.5	40.7	40.8	38.6	40.0	40.7	44.6	44.4	44.7	50.7
HRL	mean	50.5	50.1	50.5	50.1	50.1	50.1	51.3	48.5	49.4	50.9	52.0	48.4	50.6	51.6	56.0	55.5	56.1	58.5
MRL	mean	46.6	46.4	46.5	46.4	46.4	46.4	47.5	44.9	45.8	47.2	48.0	44.8	46.8	47.6	51.2	50.9	51.3	53.8
LRL	mean	34.4	34.9	34.9	34.6	34.6	34.6	35.2	33.3	34.4	35.7	35.4	33.4	34.5	35.6	38.5	38.6	38.8	50.3
URL	mean	28.7	30.2	29.5	28.8	29.5	29.1	29.2	27.8	29.9	30.2	29.2	29.2	29.7	29.5	34.8	34.5	34.5	40.1
OOM	mean	6.2	12.6	10.9	13.2	12.3	12.5	13.7	9.7	12.2	12.2	14.2	8.8	14.8	10.6	11.0	12.9	13.0	-
Δ_{UR}	mean	30.4	31.6	31.1	30.6	31.0	30.8	31.0	29.4	31.3	31.9	31.1	30.4	31.2	31.3	35.9	35.7	35.8	43.2
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
ace	URL	27.4	29.9	25.8	25.9	29.8	28.4	31.3	28.4	30.3	31.0	28.5	28.2	29.6	29.6	33.2	33.2	32.7	31.5
acm	URL	42.0	41.5	42.5	41.3	41.3	41.7	43.4	39.8	41.3	42.9	44.4	39.3	42.5	43.4	49.0	48.2	48.8	42.1
acq	URL	39.5	39.9	40.4	39.6	40.3	40.8	44.1	38.5	41.1	43.5	45.3	38.2	42.7	42.5	49.0	46.7	49.2	26.8
aeb	URL	36.4	36.4	35.9	35.1	36.1	37.6	39.4	35.4	37.2	39.1	40.5	34.0	38.4	38.5	42.4	41.2	43.6	61.2
apc	URL	40.8	40.1	40.2	39.8	39.5	39.8	43.7	35.3	41.5	42.8	45.2	37.4	41.8	42.2	50.0	47.8	50.2	38.9
ar-MA	URL	36.0	35.7	36.2	35.6	35.7	35.5	36.8	34.5	35.2	36.3	37.3	34.0	36.0	36.5	39.9	39.8	40.2	28.3
arz	URL	36.4	33.5	32.8	33.3	35.3	32.5	38.4	27.5	35.7	37.5	38.6	30.7	33.9	37.2	43.7	42.5	43.5	50.5
awa	URL	43.0	43.5	42.9	42.2	42.1	41.4	43.5	37.0	42.4	44.0	44.8	41.0	43.1	44.4	46.0	43.4	46.7	27.1
ber	URL	9.2	9.9	9.0	5.7	8.8	8.1	6.7	9.7	8.9	7.9	9.6	9.8	10.2	9.7	13.6	13.5	12.1	34.3
bm	URL	15.0	20.5	21.0	20.4	18.1	20.0	14.4	16.6	20.3	22.6	13.8	16.9	14.3	18.4	23.1	26.5	28.7	44.3
bug	URL	29.3	29.1	29.4	29.2	29.2	29.4	29.6	29.0	29.3	29.5	29.7	29.1	29.4	29.6	30.1	29.8	30.1	30.4
cjk	URL	14.1	15.9	15.9	15.6	15.8	15.0	12.9	15.3	15.5	14.7	15.6	16.2	17.3	14.4	17.3	17.1	16.1	47.3
din	URL	12.8	9.9	11.2	9.8	9.4	9.7	11.6	8.1	13.3	11.5	8.3	11.0	11.2	8.3	12.6	18.6	15.5	19.8
dyu	URL	12.8	15.6	14.4	14.3	15.0	13.8	11.7	14.5	15.9	16.2	12.3	14.8	14.4	15.1	17.7	19.6	20.6	50.7
fon	URL	12.3	13.7	12.3	11.1	13.0	11.8	10.7	12.7	12.5	10.6	11.5	14.3	13.4	10.0	18.7	17.0	15.8	31.3
hne	URL	45.8	46.1	46.1	45.6	46.7	44.0	46.2	40.2	44.6	45.4	43.1	45.0	45.3	47.6	49.1	50.0	49.3	38.5
ilo	URL	35.5	43.0	41.5	42.4	40.9	42.6	39.5	39.2	41.3	43.5	36.3	37.7	37.9	41.9	42.7	48.0	47.9	33.0
kac	URL	16.5	17.2	16.3	13.5	17.6	15.7	12.9	15.1	16.1	16.4	15.3	20.1	18.0	12.0	26.5	23.8	22.5	46.8
kbp	URL	12.6	13.4	13.6	10.2	11.4	14.1	13.0	12.4	14.2	12.4	13.0	16.2	14.6	13.7	21.2	18.3	16.2	48.4
kmb	URL	19.2	23.4	19.4	19.2	20.6	17.6	15.7	19.8	22.2	19.0	19.1	22.4	21.8	17.5	25.0	24.7	20.5	41.4
ltg	URL	29.1	29.2	29.1	29.8	30.9	30.4	31.7	30.1	30.6	31.7	30.5	31.0	30.6	31.4	38.1	38.2	38.7	34.5
lus	URL	18.0	24.2	24.2	18.1	21.8	26.6	21.6	20.3	26.2	25.7	19.5	19.0	20.0	22.4	29.3	30.5	32.8	57.5
mag	URL	47.3	48.0	46.4	46.0	47.1	45.0	48.2	38.5	45.1	48.4	47.9	45.8	47.5	49.1	47.5	50.2	51.3	23.5
mai	URL	43.2	44.4	44.0	43.9	44.1	42.0	42.9	39.6	41.5	43.8	41.3	41.4	41.9	44.4	48.8	47.7	47.9	48.2
min	URL	38.2	39.4	39.5	39.8	39.7	39.4	40.0	40.0	40.0	40.1	39.8	39.7	39.5	39.4	45.7	45.5	43.6	39.1
mni	URL	13.2	12.8	15.7	14.2	14.6	14.1	12.8	15.1	14.5	12.8	15.7	16.2	17.2	14.6	22.9	20.3	15.3	38.0
nus	URL	9.6	16.4	12.0	12.0	11.1	10.9	10.8	10.1	13.3	11.3	11.4	12.1	11.6	9.4	17.1	17.6	16.3	28.3
pag	URL	35.9	36.9	36.3	35.7	36.1	36.3	35.1	35.8	37.3	36.4	34.9	36.8	37.0	35.6	39.9	40.2	40.9	55.6
pap	URL	40.1	41.0	39.6	40.9	41.1	39.1	40.5	38.9	40.6	42.1	38.5	39.4	39.5	39.8	44.4	44.2	45.3	42.0
quy	URL	26.6	30.5	27.5	29.6	28.5	31.1	28.8	28.3	30.7	31.2	27.6	28.1	28.6	26.3	31.5	31.6	34.5	49.6
scn	URL	38.2	39.4	39.3	39.0	38.1	38.9	38.1	37.8	39.7	39.5	38.4	38.3	38.6	39.1	43.7	44.7	43.3	45.2
sg	URL	19.1	25.1	23.1	21.9	22.4	21.4	19.0	22.3	25.9	21.5	23.4	25.6	25.9	19.8	31.2	29.8	28.7	56.8
shn	URL	14.2	15.6	14.7	13.5	15.6	10.6	11.9	12.2	8.5	12.6	8.1	18.1	12.2	13.9	25.1	16.2	21.3	38.3
vec	URL	42.3	42.2	42.2	42.1	42.0	42.1	42.8	42.0	42.3	43.0	42.5	41.5	42.6	42.8	44.4	44.1	44.3	41.9
war	URL	51.5	53.2	53.0	52.9	53.1	52.3	51.7	53.2	51.2	51.2	52.3	52.0	52.3	51.9	57.1	56.1	53.1	31.8
am	LRL	30.6	30.5	31.6	30.3	30.9	30.6	32.5	28.3	29.4	32.0	33.2	28.2	31.3	32.6	38.4	37.7	38.0	59.3
as	LRL	13.1	15.3	17.0	12.9	16.0	16.0	11.1	15.5	14.5	18.6	15.9	12.8	15.1	15.9	20.7	18.2	17.2	48.9
ay	LRL	22.5	25.9	23.1	24.0	22.9	23.3	22.5	23.2	26.0	22.9	22.6	24.4	22.7	24.6	22.9	27.0	23.5	50.7
ba	LRL	14.6	16.8	13.0	14.3	15.5	15.9	13.6	15.2	13.2	17.3	16.3	17.2	12.9	15.8	16.9	15.6	18.3	50.8
ban	LRL	41.0	40.6	40.6	40.3	40.8	40.7	41.6	39.8	40.2	41.3	41.0	39.0	40.6	41.3	42.4	43.1	42.2	53.2
be	LRL	37.7	37.3	38.0	37.5	37.5	37.2	38.4	36.2	37.1	38.0	38.9	36.1	37.8	38.4	42.0	41.8	42.3	44.7
bho	LRL	14.7	17.9	20.0	21.9	24.7	19.2	23.2	18.2	23.1	25.5	24.4	19.1	16.0	21.6	21.0	15.4	22.1	51.0
bo	LRL	13.2	16.3	16.1	15.7	14.7	15.1	18.2	11.1	13.3	18.0	17.9	12.7	13.5	21.3	24.8	23.3	24.4	55.8
cy	LRL	64.4	63.9	64.7	64.1	63.8	64.0	65.8	61.4	63.1	65.5	66.8	61.2	64.3	66.1	72.2	71.9	72.6	45.4
dz	LRL	5.6	5.1	4.9	5.0	4.5	5.5	6.6	6.0	4.9	4.9	4.9	6.7	5.1	4.8	5.7	5.2	6.4	53.3
ee	LRL	26.9	29.5	28.9	28.3	27.2	29.8	28.4	25.2	28.5	30.5	26.8	26.3	26.8	28.6	28.6	32.0	31.3	40.4
eo	LRL	57.6	57.5	57.7	57.4	57.3	57.4	58.1	56.7	57.0	58.0	58.2	56.5	57.6	58.0	59.0	58.8	59.1	27.9
eu	LRL	47.6	47.1	47.8	47.0	46.8	47.0	48.4	45.4	46.8	48.1	48.8	45.4	47.7	48.4	52.6	52.4	52.4	54.8
fa-AF	LRL	47.4	47.1	47.5	46.7	47.0	46.6	48.0	45.9	46.1	47.0	49.4	45.0	48.3	48.6	53.3	53.1	52.9	54.0
ff	LRL	19.1	18.9	19.0	19.4	18.3	19.3	18.8	18.8	18.8	19.6	19.0	18.6	19.6	19.2	19.8	20.1	20.2	46.2
fj	LRL	33.7	34.8	35.6	33.7	35.6	34.0	34.0	33.7	35.5	34.8	34.0	34.2	35.7	35.2	39.7	39.6	39.7	42.3
fo	LRL	38.5	38.2	38.4	38.0	38.3	38.2	40.1	37.7	39.0	39.6	38.7	37.5	37.0	38.5	44.4	44.7	45.1	43.3
ga	LRL	47.9	47.6	48.6	47.9	47.6	47.9	49.5	46.1	47.3	49.1	50.2	45.7	48.0	49.7	55.0	54.8	55.7	40.8
gn	LRL	19.1	19.6	19.2	18.5	19.4	18.7	20.3	18.4	19.2	19.4	19.0	19.3	19.3	18.0	20.8	20.7	20.1	31.6
gu	LRL	49.8	49.5	49.9	49.1	49.4	49.4	50.7	47.7	48.6	50.3	51.2	48.0	49.9	51.0	54.1	53.6	54.5	51.1
ht	LRL	49.6	49.1	49.6	49.3	49.4	49.4	49.9	48.6	48.9	49.7	50.2	48.4	49.6	49.9	51.2	51.1	51.0	48.3
jv	LRL	51.7	51.8	52.7	52.1	51.5	51.7	52.5	50.7	51.1	52.3	53.1	50.7	51.6	52.8	54.5	54.3	54.4	50.6
kg	LRL	33.1	31.9	33.0	32.9	31.0	32.9	32.3	31.9	32.9	32.8	32.9	32.9	33.1	32.1	32.8	35.2	35.1	61.1
kn	LRL	50.5	50.4	51.0	50.0	49.9	49.9	51.4	48.4	49.2	51.2	52.2	48.4	50.6	51.7	55.5	55.0	55.7	42.8
ks	LRL	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.7	0.8	0.9	0.8	0.8	0.8	0.8	0.8	0.8	1.0	40.9
ku	LRL	33.2	33.1	32.8	32.5	32.9	32.4	33.8	31.5										

lang	cat.	B	T	T75	T50	T25	T _{Gatitos}	C _M	C _P	C _{MP}	C _M	T	G _M	G _P	G _{MP}	G _M	T	B _{BIG}	T _{BIG}	(C _M T) _{BIG}	NLLB
tg	LRL	43.3	42.7	42.5	42.5	43.1	42.1	44.4	40.6	41.4	44.0		45.0	41.0	43.4	44.7	48.3	47.9	48.3	64.0	
ti	LRL	5.9	8.0	8.1	6.6	5.2	6.8	7.7	5.5	7.0	8.1		6.8	4.0	4.5	6.3	9.6	9.0	7.6	47.7	
tk	LRL	42.0	42.7	42.1	40.3	41.6	41.9	43.7	39.2	42.4	42.9		43.9	40.1	41.7	43.7	50.8	50.5	51.1	38.6	
tn	LRL	35.6	35.5	35.1	36.5	35.3	36.1	36.1	34.7	34.8	36.2		36.7	35.0	36.2	36.7	41.1	40.7	40.5	48.0	
tpi	LRL	25.2	25.2	25.2	25.2	25.2	25.0	25.2	25.2	25.2	25.1		25.0	24.6	25.0	25.0	25.3	25.3	25.2	56.6	
ts	LRL	33.2	37.6	35.9	36.1	35.1	37.4	35.3	31.9	34.8	37.3		31.9	32.2	32.9	34.6	33.9	41.3	40.7	40.1	
tum	LRL	32.1	32.0	31.6	32.0	32.3	32.1	31.4	31.0	31.0	30.1		31.3	30.8	30.8	31.9	32.4	32.5	31.3	45.7	
ug	LRL	36.2	37.0	36.9	35.9	36.9	35.5	38.6	32.2	35.3	39.2		38.1	32.8	37.3	38.6	47.9	47.7	47.9	58.3	
wo	LRL	13.9	15.2	14.5	14.8	14.4	13.0	11.6	12.1	14.8	14.2		14.0	14.7	14.3	14.8	21.4	20.4	20.9	54.9	
xh	LRL	47.6	47.1	47.2	47.3	47.6	47.3	48.1	46.3	46.4	47.9		48.8	46.0	47.8	48.1	52.2	51.6	52.1	48.8	
af	MRL	64.5	64.1	64.5	64.2	64.2	64.2	64.9	63.0	63.5	64.3		65.5	62.9	64.5	65.1	67.9	67.5	67.8	45.0	
az	MRL	42.1	42.0	42.1	41.7	41.9	41.8	42.6	41.1	41.3	42.2		43.2	40.8	42.1	42.6	45.2	45.2	45.1	50.4	
bg	MRL	59.3	58.8	59.3	58.8	58.7	58.6	60.0	57.1	58.4	59.6		61.2	57.0	59.5	60.6	65.6	65.2	65.8	60.6	
bn	MRL	47.1	46.7	47.2	46.5	47.0	46.5	48.0	45.1	46.1	47.8		48.8	44.7	47.1	48.2	51.9	51.3	51.8	50.9	
bs	MRL	52.9	52.7	53.3	52.8	52.6	52.7	53.9	51.3	52.4	53.4		54.8	51.4	53.1	54.4	58.4	57.8	58.5	49.2	
ca	MRL	60.7	60.4	60.8	60.4	60.4	60.5	61.4	59.4	60.3	61.2		62.0	59.1	60.8	61.6	64.8	64.7	65.3	52.5	
ceb	MRL	58.6	58.6	58.7	58.6	58.4	58.1	59.5	57.6	57.9	58.7		59.4	57.5	58.9	59.0	61.2	60.5	60.9	50.6	
ckb	MRL	31.9	35.0	34.9	33.5	31.8	35.2	35.9	28.9	33.8	37.1		31.9	30.3	31.2	36.5	43.2	44.7	44.6	61.2	
et	MRL	52.3	51.7	52.4	51.8	51.9	51.7	52.9	50.0	51.0	52.5		53.9	49.8	52.2	53.2	58.2	57.8	58.3	52.6	
fil	MRL	58.1	58.1	58.6	58.0	58.0	58.0	58.8	57.2	57.7	58.4		59.1	57.1	58.3	58.7	60.8	60.7	61.0	55.7	
gd	MRL	48.9	48.4	48.8	48.6	48.6	47.9	49.1	47.4	48.1	48.9		49.8	47.0	48.6	49.7	53.0	52.6	52.9	57.3	
gl	MRL	56.8	56.6	57.0	56.9	56.6	56.5	57.3	56.0	56.3	57.0		57.7	55.8	57.0	57.5	60.0	59.7	60.1	63.2	
ha	MRL	43.6	42.7	43.5	42.8	43.2	42.4	44.3	41.7	41.4	44.0		44.3	41.7	44.4	43.5	49.1	48.3	49.0	56.0	
hr	MRL	51.9	51.7	52.1	51.5	51.6	51.5	52.8	50.3	51.3	52.6		53.5	50.3	52.1	53.1	57.0	56.6	57.0	58.8	
hy	MRL	47.7	47.9	47.9	47.3	47.4	47.7	49.3	45.2	47.2	48.5		49.8	45.0	48.0	48.9	53.3	53.3	53.6	43.6	
ig	MRL	38.2	37.7	38.2	37.6	37.9	37.5	38.5	36.8	37.1	38.8		39.2	36.7	38.0	38.8	41.3	41.0	41.4	60.1	
is	MRL	43.8	43.6	44.2	43.4	43.6	43.7	44.7	41.8	43.0	44.5		45.7	41.7	43.8	45.3	51.0	50.4	51.1	41.8	
ka	MRL	45.2	45.3	45.4	45.0	44.9	45.0	46.3	43.5	44.3	45.6		46.5	43.0	45.2	46.0	49.5	49.3	49.6	57.8	
kk	MRL	49.6	49.5	49.7	49.5	49.4	49.4	51.0	46.9	49.1	50.3		51.6	47.0	49.9	51.1	55.6	55.2	55.5	49.7	
km	MRL	38.3	38.0	38.7	38.2	38.2	37.6	39.5	36.6	37.4	39.0		40.2	36.8	38.8	39.5	43.0	43.0	43.0	66.6	
lo	MRL	44.0	44.1	44.3	44.0	44.0	43.7	44.8	42.4	43.1	44.6		45.2	41.8	44.0	44.6	47.3	47.8	47.7	61.9	
lt	MRL	49.4	48.9	49.2	49.1	48.8	49.0	50.5	47.2	48.5	49.8		51.0	47.0	49.5	50.6	56.2	55.6	56.4	52.1	
lv	MRL	51.8	51.4	51.3	51.6	51.5	51.6	52.9	49.6	50.4	52.3		53.7	49.7	52.2	53.0	58.7	57.9	58.7	48.7	
mg	MRL	44.1	44.3	44.3	44.3	44.8	43.9	45.2	42.7	43.3	45.2		45.9	42.5	44.9	45.5	49.3	48.8	49.0	43.5	
mi	MRL	40.6	40.5	40.3	40.5	40.7	40.2	41.0	40.2	40.6	40.9		41.1	39.8	40.9	40.6	41.8	41.7	41.8	60.1	
mk	MRL	57.4	57.2	57.1	57.1	57.1	57.2	58.4	55.5	56.7	57.9		58.8	55.4	57.5	58.5	61.8	61.5	61.7	51.1	
ml	MRL	48.3	47.6	47.8	47.7	47.9	47.6	49.5	45.3	46.8	49.0		50.3	44.9	48.5	50.0	55.2	53.9	55.4	59.4	
mr	MRL	45.3	44.8	44.7	44.8	44.9	44.9	46.1	43.3	44.2	45.5		46.5	43.2	45.6	46.2	50.2	49.8	50.5	52.8	
ne	MRL	49.5	49.0	48.4	48.6	48.4	48.7	49.7	47.0	48.1	49.4		50.7	47.5	49.4	50.0	53.5	53.0	53.1	27.4	
ny	MRL	44.6	44.1	44.7	44.5	44.3	44.5	45.3	43.1	43.8	44.8		45.8	43.2	44.6	45.3	48.0	47.7	48.1	50.1	
or	MRL	40.6	39.5	39.2	39.2	39.3	39.5	41.0	37.4	38.5	40.6		41.6	37.4	40.1	41.4	47.3	46.1	47.1	61.4	
sd	MRL	44.0	43.5	43.3	43.2	43.7	43.1	43.9	42.8	42.6	43.6		44.4	42.5	43.6	44.0	46.1	46.1	46.2	54.2	
sl	MRL	50.7	49.9	50.6	50.1	50.1	50.3	51.5	48.8	49.7	51.0		52.1	48.4	50.7	51.7	56.1	56.1	56.6	47.1	
sm	MRL	49.2	48.7	48.9	48.9	48.8	49.1	49.4	48.2	48.5	49.4		49.7	47.5	49.3	49.6	52.0	51.3	51.6	54.2	
sn	MRL	43.4	43.3	43.2	43.3	43.4	43.2	44.0	42.3	42.9	43.5		44.2	42.0	43.5	43.8	46.4	46.1	46.1	56.0	
so	MRL	42.3	42.1	42.5	42.0	42.2	41.7	42.8	41.1	41.3	42.5		43.4	41.1	42.6	42.6	45.7	45.3	45.6	66.8	
sq	MRL	54.1	54.1	53.9	54.0	53.8	53.9	54.9	52.8	53.5	54.8		55.4	52.6	54.3	55.0	58.1	57.9	58.2	47.4	
sr	MRL	31.2	34.8	31.7	34.8	34.4	36.8	35.8	28.1	33.1	38.9		39.4	31.5	36.6	38.2	41.9	43.7	42.5	62.3	
st	MRL	44.6	44.5	44.3	44.6	44.4	44.3	44.9	43.5	44.1	44.7		45.3	43.4	44.8	45.2	47.2	47.1	47.1	58.8	
sw	MRL	58.3	58.1	58.2	58.3	58.0	58.1	59.2	56.6	57.4	58.6		59.8	56.3	58.4	59.5	62.4	62.2	62.2	53.5	
ta	MRL	52.9	52.5	52.0	52.4	52.4	52.4	53.9	50.3	51.8	53.5		54.4	50.0	52.9	54.2	58.5	58.2	58.5	59.6	
te	MRL	52.6	52.5	52.2	52.2	52.5	51.9	53.8	49.9	51.6	53.5		54.4	49.9	52.8	53.8	58.4	58.1	58.5	53.6	
tt	MRL	34.7	34.9	34.4	34.4	34.2	35.8	38.3	33.1	36.0	37.6		37.0	33.3	35.9	37.6	39.9	40.2	42.0	59.9	
ur	MRL	46.2	46.0	45.9	45.5	45.5	45.6	46.5	44.4	45.3	46.2		47.0	44.5	46.4	46.9	49.2	48.9	49.3	62.1	
uz	MRL	51.7	51.4	51.3	51.2	51.3	51.3	52.4	49.8	50.5	51.8		53.5	49.8	51.8	52.6	56.2	55.5	55.9	45.9	
yi	MRL	36.6	36.0	36.4	36.5	36.6	35.9	36.6	36.2	36.2	36.5		36.5	36.2	36.4	36.3	37.2	36.9	37.1	58.1	
yo	MRL	20.9	20.9	20.7	21.1	20.9	20.7	21.0	20.8	20.9	21.0		21.0	20.5	20.9	20.9	21.3	21.1	21.4	57.9	
yue	MRL	12.2	12.9	12.7	12.2	13.3	12.2	13.2	10.8	10.9	12.5		13.5	11.6	13.2	13.0	17.8	17.7	17.1	20.8	
zu	MRL	49.7	49.1	49.4	49.6	49.3	49.4	50.7	47.7	48.5	50.1		51.3	47.5	49.7	50.9	55.2	54.4	54.9	67.1	
ar	HRL	47.7	47.4	48.0	47.4	47.4	47.1	49.0	45.1	46.2	48.4		50.2	44.6	47.9	49.1	56.1	55.2	56.0	59.4	
cs	HRL	50.6	50.1	50.6	50.1	50.2	49.9	51.4	48.3	49.5	51.0		52.1	48.2	50.6	51.7	56.4	55.9	56.7	53.5	
da	HRL	62.8	62.2	62.7	62.4	62.4	62.3	63.2	61.2	61.7	62.8		64.0	60.8	62.7	63.4	67.3	66.8	67.3	63.6	
de	HRL	55.9	55.7	56.2	55.5	55.5	55.4	56.7	53.8	55.0	56.2		57.5	54.0	56.0	57.1	61.4	60.9	61.6	62.6	
el	HRL	45.1	44.8	45.5	45.0	44.9	44.8	46.4	43.1	44.1	45.9		46.9	43.0	45.5	46.3	50.9	50.4	51.2	54.3	
es	HRL	51.6	51.4	51.7	51.5	51.4	51.4	51.9	50.7	51.1	51.8		52.3	50.5	51.6	52.1	54.5	54.2	54.6	61.6	
fa	HRL	46.0	45.2	46.1	45.0	45.3	45.0	46.7	44.0	44.6	46.0		47.0	43.5	45.9	47.1	50.7	49.9	50.7	59.1	
fi	HRL	48.5	48.2	48.6	48.0	48.1	48.0	49.6	46.4	47.											

lang	cat.	B	T	T ₇₅	T ₅₀	T ₂₅	T _{Gatitos}	C _M	C _P	C _{MP}	C _M T	G _M	G _P	G _{MP}	G _M T	B _{BIG}	T _{BIG}	(C _M T) _{BIG}	NLLB
uk	HRL	49.5	49.2	49.5	49.2	49.3	49.0	50.5	47.4	48.6	49.9	51.2	47.5	49.9	50.6	55.4	55.0	55.5	65.2
vi	HRL	49.9	50.0	50.0	49.7	49.9	49.8	50.7	47.9	48.8	51.0	51.8	47.9	50.4	51.2	55.9	55.3	56.1	71.4
zh	HRL	22.3	22.1	22.1	22.2	22.0	22.0	23.0	19.9	21.2	22.7	24.1	20.2	22.6	23.4	29.6	29.3	29.9	56.0
ar-Latn	OOM	3.0	3.0	3.0	3.6	3.5	2.9	2.3	2.2	2.9	3.8	2.9	3.0	2.8	3.6	1.6	2.1	2.1	-
kr	OOM	11.1	6.3	5.3	4.3	4.8	17.5	17.4	4.2	15.9	6.7	17.0	12.8	17.9	3.9	15.3	6.0	6.7	31.5
ki	OOM	8.7	14.6	13.1	15.5	15.3	15.9	14.8	13.6	15.0	14.5	16.0	10.5	16.9	12.3	14.9	16.1	12.3	40.1
taq	OOM	9.8	16.4	15.1	18.0	18.2	18.0	18.9	14.8	15.6	16.0	17.6	12.6	18.5	14.9	16.0	18.0	16.3	25.9
nn	OOM	11.7	20.4	19.8	22.3	21.8	16.0	21.9	18.8	20.7	20.2	21.7	16.5	23.4	18.5	19.4	22.4	21.1	56.1
luo	OOM	2.4	15.2	6.5	14.8	6.8	7.9	10.3	3.9	7.2	10.9	13.7	3.6	14.8	6.6	11.4	16.2	19.8	41.7
lmo	OOM	11.6	19.6	18.4	21.6	21.1	21.0	22.5	17.9	19.6	20.1	21.5	15.8	22.4	18.1	18.8	21.4	20.5	38.7
li	OOM	11.3	20.7	19.2	23.0	22.7	20.9	22.4	19.3	20.5	20.6	22.4	16.3	23.9	18.7	17.5	22.9	19.2	51.6
fur	OOM	11.8	20.3	19.3	22.5	22.1	20.7	22.0	18.2	20.5	20.1	22.5	16.0	23.8	18.8	19.5	22.2	18.7	58.7
szl	OOM	3.8	15.4	14.6	16.9	16.4	15.3	17.1	13.9	15.0	15.2	16.8	11.9	17.4	13.8	15.1	16.7	15.7	56.7
lij	OOM	11.1	20.7	19.1	22.5	22.1	22.7	22.3	18.6	22.6	20.1	22.6	16.2	23.7	18.9	20.3	22.3	18.6	56.1
bjn	OOM	9.2	15.1	14.2	16.7	16.4	16.4	16.9	13.6	14.3	14.9	16.9	11.9	17.2	13.9	14.7	16.7	14.7	51.8
sc	OOM	11.1	19.8	18.5	22.0	21.4	21.7	21.2	17.6	20.9	19.3	22.0	16.4	22.7	18.8	19.3	21.7	19.4	58.1
ss	OOM	8.8	15.4	13.7	16.5	16.2	16.6	15.6	14.2	16.0	15.3	16.8	11.1	17.8	13.5	15.5	17.1	10.5	49.2
bem	OOM	8.5	14.7	13.0	15.8	15.5	14.2	14.9	13.5	15.2	14.4	16.2	10.7	16.9	12.6	10.3	16.5	12.6	42.3
lua	OOM	9.1	15.1	14.2	16.6	16.8	17.3	16.2	13.6	15.3	14.9	16.5	12.9	17.1	13.7	15.2	16.3	14.4	39.4
kam	OOM	3.0	15.1	5.9	13.2	11.5	8.5	11.3	4.2	6.5	10.0	10.6	4.2	14.6	6.2	11.4	15.6	18.3	28.8
min-Arab	OOM	0.7	1.0	0.3	1.2	0.8	4.4	1.4	0.6	0.9	3.5	4.8	0.4	6.3	1.0	0.3	0.3	0.3	-
ace-Arab	OOM	0.7	1.3	0.3	0.6	0.9	1.3	1.2	3.0	1.0	3.7	2.0	0.8	6.5	1.9	0.4	0.4	0.5	21.1
mos	OOM	2.7	14.7	6.5	12.7	10.2	9.5	11.5	4.1	6.9	10.0	10.7	3.5	13.8	6.7	11.1	15.6	17.7	26.4
ast	OOM	11.9	21.2	19.8	23.1	23.7	24.0	22.4	18.5	21.3	20.5	24.6	16.7	24.2	19.6	20.2	23.0	20.9	59.6
ars	OOM	0.3	11.6	13.3	13.7	12.9	5.8	17.5	2.1	15.6	12.8	19.6	5.3	12.1	10.2	0.4	4.9	11.7	54.2
kea	OOM	3.3	18.8	7.1	15.5	8.6	10.6	13.5	4.6	8.8	13.1	12.4	4.1	17.4	9.3	14.1	18.2	21.2	44.9
taq-Tfng	OOM	2.0	4.6	5.5	1.7	5.2	3.9	5.3	3.7	5.2	5.1	5.9	4.7	6.6	6.3	0.8	0.6	8.1	19.3
bjn-Arab	OOM	5.9	5.5	2.1	4.7	4.7	12.4	9.6	11.9	9.5	9.8	12.6	5.4	12.7	9.6	0.4	4.1	8.0	20.4

lang	cat.	B	T	T ₇₅	T ₅₀	T ₂₅	T _{Gatitos}	C _M	C _P	C _{MP}	C _M T	G _M	G _P	G _{MP}	G _M T	B _{BIG}	T _{BIG}	(C _M T) _{BIG}	NLLB
umb	OOM	2.2	13.7	5.7	12.0	7.2	7.6	11.0	3.4	8.9	10.9	12.5	5.2	11.0	9.6	9.8	14.9	16.3	31.2
crh-Latn	OOM	8.8	14.8	14.0	16.1	15.8	16.1	16.3	13.3	15.0	14.4	16.0	11.6	16.7	13.6	14.0	16.1	13.5	51.1
kr-Arab	OOM	4.5	2.4	0.5	1.0	0.9	8.7	6.9	8.3	7.1	4.0	8.9	4.4	8.9	2.6	1.0	0.5	1.7	12.7
zh-Hant	OOM	8.3	7.8	7.2	7.2	7.3	9.6	9.1	7.3	7.9	8.9	8.5	6.9	8.0	8.4	13.9	13.9	10.4	17.6
ajp	OOM	0.4	13.1	14.2	14.3	13.9	6.2	19.1	3.4	17.5	12.9	17.6	8.6	13.3	9.8	1.1	5.6	19.5	56.0
ks-Deva	OOM	3.6	3.5	3.9	4.4	4.4	3.5	3.6	3.3	3.5	5.3	4.0	3.5	3.7	5.6	3.3	3.1	3.7	20.4
kab	OOM	1.5	4.4	2.7	8.0	1.6	4.7	2.3	2.1	1.7	2.3	1.8	1.5	3.7	0.9	4.7	3.4	4.5	37.9
azb	OOM	0.2	0.6	14.6	2.3	5.4	0.4	0.5	0.8	0.4	0.4	0.5	0.2	0.4	0.4	0.5	0.7	0.6	27.8

Table 24: Full Results of CHRf on FLORES for the models that we trained, in the en→xx direction. See Table 23 to demystify the model abbreviations.

lang	cat.	B	T	T ₇₅	T ₅₀	T ₂₅	T _{GAT}	C _M	C _P	C _{MP}	C _M T	G _M	G _P	G _{MP}	G _M T	B _{BIG}	T _{BIG}	(C _M T) _{BIG}	NLLB
μ	mean	47.2	47.2	47.1	46.7	46.8	46.4	47.4	46.0	46.4	47.3	47.9	45.9	47.3	47.8	53.3	53.0	52.9	59.9
HRL	mean	57.2	56.9	56.9	56.8	56.8	56.5	57.7	55.6	56.5	57.5	58.4	55.5	57.2	58.1	62.6	62.0	62.6	65.7
MRL	mean	52.1	51.8	51.9	51.5	51.8	51.3	52.5	50.6	51.2	52.2	53.2	50.5	52.1	52.9	58.1	57.7	58.0	63.3
LRL	mean	43.6	43.8	43.5	43.0	43.5	42.8	43.8	42.4	42.8	43.9	44.3	42.4	43.6	44.3	50.2	50.0	49.7	60.4
URL	mean	37.0	37.5	37.2	36.7	36.6	36.4	36.9	36.7	36.2	36.9	37.0	36.4	37.5	37.1	43.4	43.3	42.3	49.2
OOM	mean	32.5	32.5	32.5	31.8	32.1	31.5	33.0	31.5	31.4	32.7	33.0	31.4	32.8	32.8	37.9	37.5	37.3	-
Δ _{UL}	URL	39.0	39.4	39.1	38.6	38.6	38.3	38.9	38.4	38.2	39.0	39.2	38.2	39.3	39.3	45.4	45.3	44.5	52.6
—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
ace	URL	29.6	30.0	30.2	29.0	29.0	29.0	29.1	30.8	29.3	28.8	27.8	30.2	31.4	29.4	36.2	35.1	34.1	41.2
acm	URL	51.2	50.7	50.9	50.5	50.5	50.2	51.5	49.0	50.0	51.5	52.2	48.9	50.8	52.2	58.1	57.7	57.4	61.4
acq	URL	52.5	51.8	52.2	51.9	52.1	51.7	52.9	50.5	51.2	52.8	54.0	50.3	52.5	53.6	59.7	58.8	58.9	32.9
aeb	URL	46.1	45.9	46.0	45.7	45.8	45.4	46.6	44.4	45.1	46.4	47.2	44.4	46.1	47.2	53.2	52.3	52.5	73.7
apc	URL	50.8	50.6	50.9	50.4	50.5	50.4	51.4	49.2	50.0	51.2	52.1	49.0	51.0	52.2	59.5	58.3	58.4	62.2
ar-MA	URL	41.6	41.6	41.6	41.2	41.5	41.0	41.9	40.1	41.3	42.1	42.6	40.1	42.1	42.7	49.0	48.2	47.7	35.8
arz	URL	48.7	48.5	48.8	48.6	48.4	48.0	49.2	47.1	48.2	48.9	49.9	47.0	48.9	49.8	55.9	54.9	54.9	64.5
awa	URL	54.3	53.8	53.8	53.5	53.8	53.5	53.7	52.6	53.2	53.9	55.2	52.7	54.1	54.9	60.5	60.5	60.3	35.4
ber	URL	23.7	23.1	22.3	21.9	22.5	17.5	21.4	22.3	21.1	20.5	22.0	23.4	22.5	20.9	34.3	32.4	28.3	43.0
bm	URL	22.1	25.7	24.9	23.8	22.8	23.9	23.4	22.2	23.3	25.4	21.6	22.6	22.6	23.4	24.2	29.7	26.2	53.5
bug	URL	24.8	25.5	25.5	24.4	25.0	24.5	26.1	25.4	24.5	25.1	26.2	24.4	26.6	25.2	29.3	29.0	27.8	35.7
cjk	URL	20.2	20.7	20.6	19.6	19.8	19.1	20.5	19.9	20.3	20.9	20.1	19.8	20.6	20.2	24.1	23.7	23.7	51.3
din	URL	21.8	21.3	21.7	20.2	20.1	20.3	21.1	20.9	20.1	20.7	20.5	21.2	20.8	20.5	22.3	22.2	22.1	30.0
dyu	URL	18.2	19.1	19.4	18.2	17.7	17.8	19.2	17.4	18.0	19.1	18.6	17.6	18.5	18.3	20.3	20.4	19.3	67.4
fon	URL	19.7	21.5	20.3	19.4	19.0	19.4	19.8	20.9	19.3	19.5	19.8	20.6	21.2	19.8	22.3	23.1	20.6	36.6
hne	URL	55.4	54.4	55.0	54.6	55.2	54.8	54.3	54.5	53.7	54.7	56.0	54.0	56.0	56.4	64.0	63.9	63.9	44.9
ilo	URL	44.7	46.6	46.0	45.8	44.5	45.2	44.7	45.1	44.1	45.7	43.5	44.4	44.4	45.8	55.2	55.1	55.6	42.2
kac	URL	18.7	19.1	17.5	17.7	17.3	18.4	17.8	18.6	17.4	18.9	18.3	20.2	19.4	18.0	21.8	20.7	18.9	66.7
kbp	URL	23.5	25.0	23.4	23.1	22.8	22.9	22.9	24.9	23.3	22.9	23.4	25.0	23.6	23.6	24.5	25.8	23.2	62.3
kmb	URL	20.5	20.5	20.2	18.8	19.7	19.0	20.4	20.0	20.1	20.4	20.3	20.4	20.7	20.3	23.6	23.4	23.2	53.9
ltg	URL	43.9	44.7	44.7	44.5	44.8	44.0	43.7	44.3	43.0	43.3	44.0	43.8	45.0	44.5	55.9	58.0	57.1	33.7
lus	URL	20.3	22.9	22.9	19.9	20.9	23.4	22.7	21.7	22.2	25.2	21.5	19.8	23.2	21.4	23.5	23.7	22.6	73.8
mag	URL	56.6	55.2	55.5	56.0	55.7	55.3	56.0	55.2	54.7	55.9	57.5	54.9	57.0	57.6	63.8	63.3	63.5	35.4
mai	URL	54.8	54.4	54.7	54.7	54.5	54.5	54.6	54.0	53.7	54.9	56.8	53.8	55.7	56.1	63.2	62.0	63.4	44.6
min	URL	39.8	40.8	41.0	40.8	40.9	40.2	39.6	41.8	40.3	38.9	38.2	42.0	42.4	39.3	48.2	49.2	47.5	43.7
mni	URL	32.8	31.6	32.1	31.8	28.6	31.7	28.6	34.9	27.7	28.4	29.7	29.8	32.8	28.2	37.9	44.1	40.2	36.7
nus	URL	18.0	19.0	18.2	17.8	17.3	17.4	17.9	18.2	17.2	16.4	17.3	18.0	18.1	16.5	20.1	20.4	17.1	32.7
pag	URL	38.2	39.8	39.0	39.5	37.3	37.2	37.7	39.5	37.5	37.5	38.5	38.5	39.7	38.2	47.3	46.8	44.9	64.0
pap	URL	57.3	56.9	56.9	56.0	56.2	55.4	57.3	54.7	55.8	56.2	56.3	55.3	56.8	56.5	68.0	66.2	66.3	51.5
quy	URL	27.8	29.3	28.3	28.3	27.7	28.8	28.6	28.5	28.7	29.6	26.8	28.2	27.9	28.7	33.3	33.2	33.5	52.3
scn	URL	51.2	51.6	51.4	50.9	51.2	50.5	51.6	50.3	50.2	51.2	51.8	50.3	51.3	51.6	59.2	58.9	57.8	60.7
sg	URL	21.4	23.3	22.0	21.4	21.6	21.0	21.6	22.4	21.8	22.3	22.3	22.6	22.8	21.8	27.3	25.6	24.8	65.3

lang	cat.	B	T	T ₇₅	T ₅₀	T ₂₅	T _{CAT}	C _M	C _P	C _{MP}	C _M T	G _M	G _P	G _{MP}	G _M T	B _{BIG}	T _{BIG}	(C _M T) _{BIG}	NLLB
shn	URL	33.1	34.4	32.4	32.7	31.6	29.7	30.6	32.7	32.7	30.2	31.6	32.1	33.6	32.8	42.7	40.7	38.0	48.4
vec	URL	53.4	54.6	54.0	53.9	54.5	53.9	54.1	52.9	52.9	53.5	54.1	52.5	53.8	53.8	63.0	62.7	61.4	42.0
war	URL	58.1	58.9	58.5	58.6	58.6	58.7	57.3	57.6	56.8	57.0	58.1	57.4	58.6	57.3	66.0	65.8	65.2	42.9
am	LRL	46.8	46.5	46.9	46.2	47.1	46.1	47.4	45.2	46.0	47.0	48.8	44.7	47.3	48.5	54.7	53.8	54.2	71.1
as	LRL	41.1	41.0	38.5	37.4	39.7	37.0	38.8	36.3	36.2	41.9	41.8	37.4	39.3	41.2	49.5	47.3	46.3	45.4
ay	LRL	22.2	23.8	24.3	23.0	22.4	23.2	23.5	22.9	23.6	24.6	21.8	22.5	23.1	23.4	25.7	28.8	28.7	70.2
ba	LRL	37.7	36.0	32.9	35.1	35.9	31.0	37.0	34.1	33.5	40.2	39.4	35.6	35.2	38.7	41.4	39.8	42.1	52.1
ban	LRL	44.9	44.8	45.1	43.6	44.1	44.0	44.3	45.1	43.5	43.6	44.3	44.9	45.2	45.7	48.9	49.9	48.5	54.9
be	LRL	47.9	47.6	47.6	47.3	47.4	47.1	48.2	46.5	47.2	48.0	48.8	46.2	47.7	48.5	52.2	51.8	52.3	48.2
bho	LRL	47.1	46.7	46.9	46.6	47.1	46.8	47.1	46.5	46.4	47.1	48.2	46.3	47.8	48.2	54.5	54.1	54.1	53.4
bo	LRL	13.7	15.2	12.4	14.4	13.7	14.0	16.2	13.0	16.0	17.2	16.2	14.5	15.6	16.3	22.2	20.7	19.5	63.6
cy	LRL	68.5	68.2	68.5	67.7	68.1	67.5	69.2	66.2	67.3	69.0	70.0	65.8	68.5	69.6	75.3	74.9	75.6	44.0
dz	LRL	21.4	23.3	21.5	18.2	22.2	18.5	21.3	20.3	22.0	22.4	21.6	20.0	20.5	20.9	29.9	29.2	28.8	63.0
ee	LRL	27.4	28.9	28.8	27.6	26.8	28.5	27.4	28.4	27.7	28.7	26.7	29.3	27.6	28.8	28.0	30.0	26.7	52.0
eo	LRL	61.7	61.3	61.4	61.2	61.3	61.1	62.0	60.4	61.1	61.9	62.4	60.3	61.6	62.4	65.8	65.2	66.0	52.4
eu	LRL	50.7	50.2	50.2	50.1	50.1	49.7	51.1	48.6	49.9	50.7	51.8	48.8	50.5	51.7	56.9	56.3	56.9	68.0
fa-AF	LRL	54.6	54.1	54.3	54.0	54.4	53.8	55.0	52.9	53.4	54.7	55.9	53.1	54.6	55.7	61.2	60.2	60.9	61.7
ff	LRL	20.5	21.9	21.6	20.9	20.4	20.7	21.3	20.5	21.5	22.0	20.4	20.9	21.0	20.9	22.8	23.8	23.6	61.2
fj	LRL	32.5	31.6	31.3	29.7	31.2	29.4	29.7	32.8	28.9	28.8	31.7	32.4	32.1	29.3	36.5	36.7	32.5	54.6
fo	LRL	50.6	50.5	50.3	50.4	50.3	49.8	51.3	49.7	50.3	50.6	51.4	49.3	50.4	50.9	58.9	58.2	58.6	52.1
ga	LRL	57.0	56.6	56.3	56.3	56.2	56.2	57.5	55.0	56.1	57.5	58.5	54.8	57.1	58.0	65.2	64.4	65.4	61.8
gn	LRL	31.6	34.3	33.8	32.6	32.5	32.8	32.4	31.2	31.4	34.4	30.6	31.6	31.9	33.6	42.2	42.4	42.3	52.9
gu	LRL	56.6	56.0	56.1	55.9	56.3	55.8	57.1	54.7	55.6	56.8	57.8	54.7	56.6	57.3	63.0	62.3	63.1	60.1
ht	LRL	55.3	54.8	55.0	55.1	55.0	54.4	55.7	54.0	54.5	55.5	56.5	54.0	55.2	56.1	61.3	60.7	61.2	63.6
jv	LRL	52.2	51.5	52.0	51.4	51.6	51.3	52.5	51.1	51.1	52.3	53.3	50.8	52.3	53.1	58.5	57.9	57.9	54.6
kg	LRL	30.9	31.8	31.6	31.4	31.0	31.0	30.4	32.6	30.9	30.5	30.5	32.5	31.3	31.3	34.2	35.1	31.1	77.2
kn	LRL	53.1	52.7	52.7	52.7	52.9	52.4	53.6	51.4	52.0	53.1	54.4	51.2	53.2	53.9	59.0	58.4	59.2	54.6
ks	LRL	31.6	31.5	32.6	30.5	31.6	30.0	32.2	31.1	30.9	32.4	33.7	32.0	33.2	34.1	43.0	43.2	40.1	43.7
ku	LRL	43.5	43.1	43.4	42.8	42.9	42.3	43.4	42.2	42.4	43.6	44.3	42.1	43.5	44.1	51.2	49.8	49.9	65.0
ky	LRL	45.3	45.0	44.9	44.9	45.2	44.3	45.8	43.6	44.6	45.7	46.5	43.6	45.5	46.2	50.3	50.2	50.2	64.0
lb	LRL	59.8	59.2	59.3	59.0	59.3	59.0	60.3	58.2	58.6	60.1	61.1	58.1	59.9	60.6	66.5	66.2	66.6	65.4
lg	LRL	29.6	31.0	30.5	29.5	29.2	29.9	29.9	30.2	30.0	31.0	29.1	29.9	29.3	30.3	38.7	38.1	36.2	73.4
ln	LRL	36.5	37.5	37.3	36.2	36.3	36.7	36.6	35.6	36.4	37.7	35.7	35.5	36.0	37.9	44.9	44.4	44.3	59.5
mn	LRL	47.8	47.5	47.4	47.1	47.2	46.9	48.2	46.0	46.8	47.8	49.2	45.9	48.0	48.9	54.3	54.1	54.3	62.7
mt	LRL	69.7	69.1	69.2	69.2	69.3	68.9	70.2	68.0	68.6	69.8	70.9	67.8	69.5	70.6	75.5	75.0	75.4	68.1
my	LRL	44.7	43.9	44.0	43.7	44.5	43.0	45.5	42.4	44.0	45.0	45.6	42.6	44.3	45.8	50.5	49.9	50.4	66.7
nso	LRL	47.8	47.9	47.9	47.2	47.1	47.1	47.6	46.2	46.9	47.5	48.4	46.5	47.8	48.7	57.1	56.6	56.2	52.5
oc	LRL	65.8	65.8	65.6	65.1	65.3	65.2	66.6	64.6	66.0	66.5	66.3	64.6	66.0	66.4	71.2	70.4	70.3	67.9
om	LRL	30.3	31.3	30.5	28.5	30.3	29.7	29.4	29.9	29.0	30.6	28.4	29.8	28.9	30.0	43.5	42.3	42.1	61.8
pa	LRL	55.6	55.1	55.5	55.4	55.4	54.9	56.2	54.0	54.9	56.3	57.2	53.8	56.1	56.9	63.6	62.5	63.2	59.8
ps	LRL	47.8	46.8	47.2	46.8	46.9	46.7	47.8	46.3	46.5	47.8	48.5	45.8	47.7	48.1	54.0	53.2	52.8	71.2
rn	LRL	36.0	35.5	35.4	35.3	35.7	34.9	34.6	34.7	34.6	34.7	36.1	35.1	35.5	35.8	44.4	43.1	43.1	57.4
rw	LRL	41.1	40.4	40.1	39.4	40.2	39.3	39.5	39.5	39.2	39.3	40.8	39.6	40.6	41.0	49.9	48.7	49.0	65.2
sa	LRL	39.3	40.0	39.8	39.8	39.8	40.1	40.6	39.5	40.1	40.5	40.2	39.1	39.7	40.6	46.6	46.7	47.5	77.1
si	LRL	50.7	50.4	50.0	50.3	50.2	49.9	51.6	48.6	49.8	51.1	52.7	48.4	50.7	52.2	58.3	57.2	58.3	77.6
su	LRL	52.9	52.2	52.7	52.4	52.6	52.2	53.1	51.7	51.8	52.9	53.6	51.4	53.1	53.3	57.7	57.4	57.0	61.1
tg	LRL	50.3	50.1	49.6	49.8	49.9	49.2	50.6	48.2	49.4	50.5	51.7	48.2	50.3	51.3	57.8	56.8	57.4	70.2
ti	LRL	30.2	31.3	30.7	27.3	33.2	28.5	31.5	26.0	29.4	31.1	32.3	24.5	30.4	31.3	42.1	40.5	40.1	56.3
tk	LRL	50.6	50.0	50.1	50.0	49.8	49.3	51.0	48.0	49.2	50.4	52.2	48.4	50.6	51.5	59.0	58.2	58.6	40.9
tn	LRL	43.9	43.8	43.6	43.5	43.9	43.1	43.6	42.4	42.6	43.4	44.6	42.4	43.8	44.2	50.0	49.7	49.6	63.7
tpi	LRL	44.1	46.0	45.3	46.3	45.7	44.7	43.0	43.8	42.6	42.2	41.6	43.9	44.2	43.4	41.0	50.1	46.1	68.4
ts	LRL	35.7	38.8	38.4	37.6	37.8	38.7	35.7	36.3	36.4	37.2	36.0	37.0	36.4	37.2	41.1	44.6	41.2	58.4
tum	LRL	32.1	34.1	33.1	31.9	32.7	32.7	31.8	33.2	32.5	32.3	32.7	33.2	32.6	32.8	34.7	34.4	34.3	55.4
ug	LRL	43.2	43.2	43.2	42.8	43.0	42.3	43.9	41.7	42.6	43.6	44.5	41.4	43.2	44.5	49.9	49.1	49.3	63.1
wo	LRL	31.3	31.6	31.0	31.0	31.7	31.1	29.6	28.8	29.7	29.0	29.5	29.6	30.3	29.3	40.8	40.3	37.7	61.2
xh	LRL	49.4	48.9	48.9	48.8	49.0	48.6	49.5	47.8	48.2	48.9	50.1	47.7	49.4	49.4	55.8	55.4	54.9	51.9
af	MRL	70.4	69.9	70.2	69.9	69.9	69.7	70.7	69.2	69.6	70.4	71.7	68.8	70.4	71.2	75.1	74.5	74.9	60.6
az	MRL	48.0	47.6	47.9	47.6	47.9	47.2	48.5	46.5	47.2	48.2	49.2	46.4	48.0	48.7	52.9	52.5	52.5	66.2
bg	MRL	61.4	60.8	61.0	61.0	61.2	60.8	61.8	59.8	60.5	61.6	62.3	59.8	61.3	62.1	66.3	65.9	66.5	68.4
bn	MRL	53.2	52.7	53.1	52.6	52.7	52.2	54.2	51.3	52.5	53.3	54.5	51.1	53.3	54.3	59.8	59.4	60.0	58.3
bs	MRL	60.8	60.5	60.6	60.5	60.3	60.4	61.5	59.6	60.2	61.2	62.0	59.3	60.9	61.7	66.0	65.6	65.8	62.9
ca	MRL	64.3	63.9	64.2	64.0	63.9	63.8	64.7	63.2	63.8	64.6	65.2	63.1	64.4	65.1	68.6	68.1	68.5	61.6
ceb	MRL	58.5	58.2	58.2	58.0	58.3	57.7	58.1	57.8	56.9	57.8	59.3	57.2	58.4	58.7	65.3	64.7	64.6	58.3
ckb	MRL	42.8	44.0	43.8	43.2	42.3	43.3	43.4	41.9	42.0	44.2	43.2	41.3	42.2	44.5	55.1	54.0	54.2	67.4
et	MRL	56.7	56.0	56.0	56.0	56.1	55.6	57.0	54.9	55.3	56.6	57.8	54.6	56.5	57.3	63.1	62.4	63.0	59.5
fil	MRL	60.9	60.8	60.8	60.6	60.5	60.3	61.6	59.6	59.9	61.1	62.4	59.5	61.0	61.9	67.5	66.8	67.3	61.6
gd	MRL	51.8	51.2	50.9	50.6	51.1	50.7	52.0	50.0	51.1	51.8	52.8	49.7	51.3	52.2	57.8	57.5	58.1	64.9
gl	MRL	61.5	61.4	61.3	61.2	61.3	61.2	61.8	60.5	61.1	61.8	62.2	60.5	61.6	62.3	65.1			

lang	cat.	B	T	T ₇₅	T ₅₀	T ₂₅	T _{GAT}	C _M	C _P	C _{MP}	C _M T	G _M	G _P	G _{MP}	G _M T	B _{BIG}	T _{BIG}	(C _M T) _{BIG}	NLLB
or	MRL	50.3	49.5	49.5	49.2	49.3	48.7	50.5	48.2	49.3	50.0	52.0	47.9	50.2	51.2	60.3	59.1	59.9	68.8
sd	MRL	53.1	52.3	52.7	52.3	52.3	52.1	53.3	51.7	52.1	53.2	54.1	51.2	53.3	53.6	60.0	59.1	59.3	59.1
sl	MRL	56.0	55.8	55.7	55.6	55.6	55.2	56.2	54.5	55.1	56.0	57.0	54.5	55.9	56.6	61.1	60.7	61.1	56.5
sm	MRL	46.0	46.5	46.6	45.5	46.5	45.4	46.1	45.3	45.7	45.8	47.3	45.3	46.3	46.7	51.6	51.6	49.6	64.2
sn	MRL	42.3	42.0	42.0	41.7	42.0	41.2	42.1	41.5	41.4	42.0	43.1	41.4	42.2	42.4	47.3	46.7	46.4	58.7
so	MRL	41.1	40.8	41.1	40.0	41.2	40.1	41.3	40.5	40.3	40.7	42.1	40.6	41.7	41.2	46.7	46.6	46.1	71.1
sq	MRL	59.9	59.6	59.5	59.4	59.7	59.2	60.3	58.5	59.0	59.8	61.0	58.0	59.9	60.5	65.3	64.8	65.2	54.4
sr	MRL	60.2	59.9	59.7	60.1	59.8	59.6	60.7	58.6	59.1	60.5	61.6	58.3	60.3	61.0	66.3	66.1	66.3	71.3
st	MRL	50.3	49.7	49.9	49.6	49.7	49.3	50.2	48.7	48.9	49.9	51.2	48.7	49.9	50.8	58.1	57.1	57.4	63.0
sw	MRL	56.3	55.8	55.9	55.6	55.9	55.5	56.6	54.4	54.8	56.0	57.7	54.3	56.1	57.1	63.2	62.2	62.6	63.4
ta	MRL	51.3	50.6	50.9	50.4	50.7	50.1	51.8	49.0	50.2	51.2	52.3	49.2	51.1	51.9	57.8	56.8	57.6	66.8
te	MRL	56.6	55.3	55.7	55.6	55.6	55.2	56.7	53.6	55.0	56.1	57.6	53.9	56.1	57.3	62.8	61.9	62.7	60.9
tt	MRL	48.3	48.2	48.0	48.2	48.3	47.7	48.5	46.7	47.6	48.3	49.0	46.6	47.9	48.9	55.9	55.5	55.5	64.6
ur	MRL	53.0	52.5	52.7	52.4	52.8	52.3	53.7	51.5	52.2	53.4	54.6	51.4	53.2	54.2	59.5	58.7	59.5	69.4
uz	MRL	53.3	52.7	52.9	52.6	52.8	52.1	53.9	51.4	52.3	53.2	54.7	51.3	53.4	54.2	59.6	59.4	59.6	54.1
yi	MRL	48.2	47.5	49.0	45.9	46.7	47.9	48.7	45.3	45.9	48.7	48.0	45.2	46.3	48.4	52.1	53.9	51.6	61.2
yo	MRL	35.4	35.5	35.4	35.0	35.2	34.8	35.8	34.0	35.2	35.8	36.5	34.1	35.2	36.0	40.6	40.1	40.4	62.9
yue	MRL	46.1	46.6	46.4	46.6	46.5	45.9	46.6	44.9	45.3	46.6	47.3	44.2	46.2	47.4	54.4	53.9	54.3	58.5
zu	MRL	49.9	49.2	49.3	48.9	49.5	48.7	49.7	48.3	48.2	50.2	50.7	48.1	49.9	51.1	56.3	56.4	56.3	68.3
ar	HRL	55.7	55.2	55.5	55.2	55.1	55.1	56.5	53.5	54.4	56.0	57.3	53.4	55.5	57.2	63.1	62.4	62.9	63.0
cs	HRL	59.0	58.7	58.9	58.7	58.6	58.5	59.5	57.6	58.5	59.4	60.2	57.4	59.0	60.0	64.0	63.4	63.9	65.7
da	HRL	66.2	65.6	65.8	65.7	65.5	65.5	66.6	64.8	65.3	66.0	66.8	64.7	66.0	66.5	70.1	69.7	70.3	70.4
de	HRL	61.4	61.0	61.2	60.9	61.0	60.7	62.1	59.8	60.7	61.7	62.6	59.6	61.4	62.2	66.6	66.1	66.7	69.0
el	HRL	54.9	54.6	54.6	54.8	54.8	54.5	55.5	53.5	54.6	55.4	56.3	53.4	55.0	56.0	60.0	59.5	60.5	64.9
es	HRL	55.5	55.5	55.4	55.3	55.4	55.3	55.8	54.5	55.4	55.9	56.3	54.7	55.5	56.1	59.4	59.1	59.3	66.1
fa	HRL	54.6	54.2	54.3	54.3	54.5	53.9	55.0	52.8	53.6	54.9	56.1	52.6	54.8	55.7	60.7	59.8	60.7	66.7
fi	HRL	53.0	52.7	52.6	52.5	52.2	52.1	53.7	50.9	52.2	53.5	54.6	50.9	53.0	54.0	60.0	59.2	60.0	69.7
fr	HRL	63.5	63.2	63.2	63.2	63.3	63.1	63.8	62.3	62.8	63.6	64.5	62.0	63.7	64.1	67.7	67.2	67.6	68.3
hi	HRL	59.2	58.5	58.8	58.6	58.8	58.5	59.5	57.8	58.1	59.1	60.3	57.3	59.3	60.0	64.8	64.0	64.6	71.3
hu	HRL	54.0	53.7	53.8	53.6	53.5	53.2	54.4	52.2	53.1	54.3	55.4	51.9	53.8	55.0	60.4	59.8	60.4	63.7
id	HRL	61.8	61.4	61.7	61.4	61.4	61.2	62.2	60.6	61.0	61.8	62.7	60.3	61.9	62.6	66.4	66.0	66.3	60.0
it	HRL	57.5	57.2	57.1	57.3	57.3	57.0	57.7	56.3	56.9	57.7	58.3	56.3	57.5	57.9	61.5	61.1	61.4	68.0
iw	HRL	56.4	55.9	55.9	55.5	55.8	55.5	57.6	53.8	55.6	57.0	58.4	53.7	56.3	57.7	64.3	63.6	64.8	64.2
ja	HRL	47.9	47.7	47.6	47.4	47.6	47.2	48.6	46.1	47.3	48.5	49.3	45.9	48.1	49.1	54.5	53.7	54.5	66.4
ko	HRL	48.2	48.3	48.1	47.9	48.1	47.6	49.6	46.5	48.1	49.2	49.9	46.1	48.7	49.8	55.2	54.8	55.5	68.6
ms	HRL	62.1	61.6	61.8	61.7	61.8	61.3	62.4	60.7	61.2	62.1	63.3	60.5	62.2	63.0	67.2	66.4	66.9	69.0
nl	HRL	55.2	55.1	54.9	54.8	54.8	54.7	55.4	54.1	54.6	55.2	56.0	53.9	55.0	55.6	58.6	58.5	58.8	71.2
no	HRL	62.6	62.4	62.2	62.3	62.3	62.1	63.0	61.3	61.7	62.8	63.4	61.1	62.5	63.2	66.6	65.8	66.5	59.0
pl	HRL	51.5	51.5	51.4	51.0	51.1	50.9	52.1	50.2	50.9	51.8	52.6	50.3	51.6	52.4	55.7	55.0	55.8	66.1
pt	HRL	66.3	66.0	66.0	66.0	66.0	65.8	66.6	65.2	65.9	66.5	67.4	65.2	66.5	67.0	70.3	69.7	70.1	59.5
ro	HRL	62.8	62.7	62.5	62.4	62.4	62.4	63.5	61.5	62.2	63.0	64.1	61.3	62.8	63.6	67.6	67.3	67.4	59.5
ru	HRL	56.1	56.0	56.0	55.8	55.9	55.5	56.6	54.8	55.4	56.3	57.2	54.5	56.1	56.8	60.8	60.1	60.8	72.7
sk	HRL	58.5	58.5	58.3	58.5	58.5	58.2	59.3	57.3	58.0	58.9	60.2	57.2	58.9	59.7	64.4	63.8	64.2	58.2
sv	HRL	65.2	64.8	64.9	64.7	64.9	64.5	65.5	63.7	64.4	65.2	66.1	63.4	65.0	65.6	69.7	69.4	69.6	63.3
th	HRL	49.2	48.5	48.3	48.1	48.2	47.3	49.4	46.8	48.2	49.4	50.4	47.2	48.8	49.7	56.1	55.4	56.3	61.2
tr	HRL	57.1	56.4	56.5	56.3	56.1	56.0	57.3	55.1	55.8	57.2	58.4	54.5	57.0	57.7	63.1	62.2	63.0	63.7
uk	HRL	57.9	57.4	57.3	57.3	57.4	57.3	58.2	56.4	57.0	58.0	59.1	56.1	57.8	58.5	63.3	62.6	63.3	69.2
vi	HRL	54.1	53.5	53.2	53.4	53.3	53.3	54.5	52.2	53.4	54.3	55.5	52.2	54.3	54.7	59.7	59.0	59.5	70.1
zh	HRL	48.5	48.7	48.4	48.6	48.5	48.1	49.3	46.7	48.0	49.1	50.2	46.8	49.0	50.0	55.0	54.2	55.1	61.4

Table 25: Full Results of CHRf on FLORES for the models that we trained, in the xx→en direction. See Table 23 to demystify the model abbreviations.